



INTELLIGENT WATERING SYSTEM

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**BACHELOR OF ENGINEERING
IN
COMPUTER ENGINEERING**

**SCHOOL OF INFORMATION TECHNOLOGY
MAE FAH LUANG UNIVERSITY**

2020

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**SIMON YOSBOON
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**A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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IN
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IN
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2020

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ABSTRACT

This paper presents an Intelligent watering system. This system is based on the watering system for vegetable plots that use coconut cloves instead of soil. It is a system that controls the water level in the plot instead of measuring soil moisture or scheduling with a timer in conventional watering systems. There are 4 main subsystems consisting of pH adjustment system, automatic mixing system (NPK), automatic tank water filling system and the automatic watering system. Most importantly, this system is flexible and customizable to users. The system allows the user to set up or adjust the various subsystems as the user needs.

Keywords: Arduino Mega 2560, Analog pH sensor, Electrical Conductive Sensor, pH calibration.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

Nowadays, technology has played a role in agriculture quite a lot. To facilitate farmers in cultivating various plants, For example, automatic watering systems or smart farms. The general user is usually only interested in watering for a certain time or taking care of vegetables to grow at a specific time by controlling factors such as temperature, humidity in the air, soil moisture, light, and fertilizers. The most commonly sold smart farm systems can control all of the above except fertilizers. Waste of time and resources from mixing fertilizers that do not meet the desired formula can affect the growth of plants. Therefore, accurate mixing of fertilizers is important as any other factor.

The Intelligent watering system is designed to be suitable for the precise mixing of fertilizers according to the formulation with a simple procedure and able to provide water to plants in greenhouses accurately and precisely. The system comes with an automatic NPK fertilizer mixing system, water level control system in the plot, automatic fertilizer mixing tank and adjustment system, pH adjustment system, and automatic fertilizer. The system will help reduce wasting of water and save more time for that process.

1.2 Objective

1.2.1 To develop an Intelligent watering system by using microcontroller.

1.2.2 The Intelligent watering system can work precisely.

1.3 Scope

1.3.1 User

- 1) User can select working mode (Auto / Hybrid / Manual)
- 2) User can choose operation of the subsystem.
- 3) User can choose liquid to be filled into the mixing tank.
- 4) User can set the liquid level in the mixer tank.
- 5) User can select liquid to water into plots.
- 6) User can set the water level into plots.
- 7) User can define pH value in mixing tank.
- 8) User can monitor and calibrate EC value in plots.
- 9) User can set the recipes of NPK fertilizer.
- 10) User can design their own fertilizer recipe.
- 11) User can set time of the machine.
- 12) User can test all essential devices in the system.
- 13) User can set technical constants of the system.
- 14) User can restore all setting and constant back to factory settings.

	Computer Engineering Pre-Project					Computer Engineering Project				
Plan/Month	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Proposal preparation and defense										
Literature review and requirement gathering										
System analysis										
System design										
Computer Engineering Pre-Project document preparation										
Computer Engineering Pre-Project Exam										
System development										
System testing										
Computer Engineering Project document preparation										
Computer Engineering Project Exam										

1.6 Expected Result

1.6.1 Intelligent watering system.

1.6.2 The Intelligent watering system works precisely.

1.7 Place and Equipment

1.7.1 Place

- 1) Mae Fah Luang laboratory S7B room 315
- 2) Mae Fah Luang laboratory S3 room 114
- 3) Mae Fah Luang Library
- 4) Dormitory

1.7.2 Hardware

- 1) Arduino Mega 2560
- 2) Pilot-Lamp status
- 3) ZP-OP2 Phase protection
- 4) Ultrasonic HC-SR04
- 5) Schneider RXM2LB2BD Relay 24VDC 5A
- 6) Switching Power Supply 24V 10A
- 7) MSOT 10 Mitsu MSOT10+ 5A (4-6)
- 8) Safety breaker HACO SB-N16L 10A
- 9) RCBO Consumer Unit CT ELECTRIC CHONG-2 32A
- 10) Controller container Super TSLC-2 52x35x17 cm.
- 11) Button
- 12) Prototype PCB board
- 13) Relay Module 5V 16 Channel
- 14) Real-Time Clock - DS3231 Module
- 15) 2004 LCD+I2C
- 16) Keypad
- 17) Buck-Boost convertor: XL6019,5-32VDC/4A(20W)
- 18) Analog pH sensor PE HOSE
- 19) EC sensor Grove v1.0
- 20) Solenoid valve 24VDC
- 21) Peristaltic pump 24VDC
- 22) Wiring materials.

1.7.3 Software

- 1) Arduino IDE

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

2.1.1 Fertilizer [3]

A fertilizer (American English) or fertiliser (British English) is any material of natural or synthetic origin that is applied to soil or to plant tissues to supply plant nutrients. Fertilizers may be distinct from liming materials or other non-nutrient soil amendments. Many sources of fertilizer exist, both natural and industrially produced. For most modern agricultural practices, fertilization focuses on three main macro nutrients: Nitrogen (N), Phosphorus (P), and Potassium (K) with occasional addition of supplements like rock dust for micronutrients. Farmers apply these fertilizers in a variety of ways: through dry or pelletized or liquid application processes, using large agricultural equipment or hand-tool methods.



Figure 2.1 NPK Fertilizer Macro nutrient for plant growth

Mechanism of Fertilizer

Six tomato plants grown with and without nitrate fertilizer on nutrient-poor sand/clay soil. One of the plants in the nutrient-poor soil has died. Fertilizers enhance the growth of plants. This goal is met in two ways, the traditional one being additives that provide nutrients. The second mode by which some fertilizers act is to enhance the effectiveness of the soil by modifying its water retention and aeration. This article, like many on fertilizers, emphasizes the nutritional aspect. Fertilizers typically provide, in varying proportions:

Three main macronutrients:

- Nitrogen (N): leaf growth.
- Phosphorus (P): Development of roots, flowers, seeds, fruit.
- Potassium (K): Strong stem growth, movement of water in plants, promotion of flowering and fruiting.

Three secondary macronutrients: calcium (Ca), magnesium (Mg), and sulfur(S);

Micronutrients: copper (Cu), iron (Fe), manganese (Mn), molybdenum (Mo), zinc (Zn), boron (B). Of occasional significance are silicon (Si), cobalt (Co), and vanadium (V).

NPK fertilizers

NPK fertilizers are three-component fertilizers providing nitrogen, phosphorus, and potassium. There exist two types of NPK fertilizers: compound and blends. Compound NPK fertilizers contain chemically bound ingredients, while blended NPK fertilizers are physical mixtures of single nutrient components.

NPK rating is a rating system describing the amount of nitrogen, phosphorus, and potassium in a fertilizer. NPK ratings consist of three numbers separated by dashes (e.g., 10-10-10 or 16-4-8) describing the chemical content of fertilizers. The first number represents the percentage of nitrogen in the product; the second number, P_2O_5 ; the third, K_2O . Fertilizers do not actually contain P_2O_5 or K_2O , but the system is a conventional shorthand for the amount of the phosphorus (P) or potassium (K) in a fertilizer. A 50-pound (23 kg) bag of fertilizer labeled 16-4-8 contains 8 lb (3.6 kg) of nitrogen (16% of the 50 pounds), an amount of phosphorus equivalent to that in 2 pounds of P_2O_5 (4% of 50 pounds), and 4 pounds of K_2O (8% of 50 pounds). Most fertilizers are labeled according to this N-P-K convention, although Australian convention, following an N-P-K-S system, adds a fourth number for sulfur, and uses elemental values for all values including P and K.

2.1.2 pH [4]

In chemistry, pH is a scale used to specify the acidity or basicity of an aqueous solution. Acidic solutions (solutions with higher concentrations of H^+ ions) are measured to have lower pH values than basic or alkaline solutions.



Figure 2.2 Test tubes containing solutions of pH 1–10 colored with an indicator

The pH scale is logarithmic and inversely indicates the concentration of hydrogen ions in the solution. This is because the formula used to calculate pH approximates the negative of the base 10 logarithm of the molar concentration^[a] of hydrogen ions in the solution. More precisely, pH is the negative of the base 10 logarithm of the activity of the H⁺ ion.

At 25 °C, solutions with a pH less than 7 are acidic, and solutions with a pH greater than 7 are basic. Solutions with a pH of 7 at this temperature are neutral (e.g. pure water). The neutral value of the pH depends on the temperature, being lower than 7 if the temperature increases. The pH value can be less than 0 for very strong acids, or greater than 14 for very strong bases.

The pH scale is traceable to a set of standard solutions whose pH is established by international agreement. Primary pH standard values are determined using a concentration cell with transference, by measuring the potential difference between a hydrogen electrode and a standard electrode such as the silver chloride electrode. The pH of aqueous solutions can be measured with a glass electrode and a pH meter, or a color-changing indicator. Measurements of pH are important in chemistry, agronomy, medicine, water treatment, and many other applications.

pH Measurement [5]

pH is defined as the decimal logarithm of the reciprocal of the hydrogen ion activity, a_{H^+} , in a solution.

$$pH = -\log_{10}(a_{H^+}) = \log_{10}\left(\frac{1}{a_{H^+}}\right)$$

For example, for a solution with a hydrogen ion activity of 5×10^{-6} (at that level, this is essentially the number of moles of hydrogen ions per litre of solution) there is $1/(5 \times 10^{-6}) = 2 \times 10^5$, thus such a solution has a pH of $\log_{10}(2 \times 10^5) = 5.3$. Consider the following example: a quantity of 10^7 moles of pure (pH 7) water, or 180 metric tonnes (18×10^7 g), contains close to 18 g of dissociated hydrogen ions.

Note that pH depends on temperature. For instance, at 0 °C the pH of pure water is about 7.47. At 25 °C it is 7.00, and at 100 °C it is 6.14. This definition was adopted because ion-selective electrodes, which are used to measure pH, respond to activity. Ideally, electrode potential, E , follows the Nernst equation, which, for the hydrogen ion can be written as

$$E = E^0 + \frac{RT}{F} \ln(a_{H^+}) = E^0 - \frac{2.303RT}{F} pH$$

where E is a measured potential, E^0 is the standard electrode potential, R is the gas constant, T is the temperature in kelvins, F is the Faraday constant. For H⁺ number of electrons transferred is one. It follows that electrode potential is proportional to pH when pH is defined in terms of activity. Precise measurement of pH is presented in International Standard ISO 31-8 as follows: A galvanic cell is set up to measure the electromotive force (e.m.f.) between a reference electrode and an electrode sensitive to the hydrogen ion activity when they are both immersed in the same aqueous solution. The reference electrode may be a silver chloride electrode or a calomel electrode. The hydrogen-ion selective electrode is a standard hydrogen electrode.

Reference electrode | concentrated solution of KCl || test solution | H₂ | Pt[clarification needed]

Firstly, the cell is filled with a solution of known hydrogen ion activity and the emf, E_S , is measured. Then the emf, E_X , of the same cell containing the solution of unknown pH is measured.

$$pH(X) = pH(S) + \frac{E_S - E_X}{z}$$

The difference between the two measured emf values is proportional to pH. This method of calibration avoids the need to know the standard electrode potential. The proportionality constant, $1/z$ is ideally equal to

$$\frac{1}{2.303RT/F} \text{ the "Nernstian slope".}$$

To apply this process in practice, a glass electrode is used rather than the cumbersome hydrogen electrode. A combined glass electrode has an in-built reference electrode. It is calibrated against buffer solutions of known hydrogen ion activity. IUPAC (International Union of Pure and Applied Chemistry) has proposed the use of a set of buffer solutions of known H^+ activity. Two or more buffer solutions are used in order to accommodate the fact that the "slope" may differ slightly from ideal. To implement this approach to calibration, the electrode is first immersed in a standard solution and the reading on a pH meter is adjusted to be equal to the standard buffer's value. The reading from a second standard buffer solution is then adjusted, using the "slope" control, to be equal to the pH for that solution. Further details, are given in the IUPAC recommendations. When more than two buffer solutions are used the electrode is calibrated by fitting observed pH values to a straight line with respect to standard buffer values. Commercial standard buffer solutions usually come with information on the value at 25 °C and a correction factor to be applied for other temperatures.

The pH scale is logarithmic and therefore pH is a dimensionless quantity.

2.2 Related Work

2.2.1 Automatic fertilizer mixer [2]



Figure 2.3 Automatic fertilizer mixer for the main fertilizers (N-P-K) in appropriate proportions.

Chemical fertilizers are the main cost of farming today. The use of unsuitable chemical fertilizers is too much or too little. Or not correct according to the principles of the production of farmers increased. There is a problem of soil degradation. As a result of this problem, the idea was to solve the problem by mixing the formulated chemical fertilizers or self-fertilizers from the main fertilizers (N-P-K) in appropriate proportions. Based on the analysis of the nutrient content in the soil according to the needs of each plant species. The solution to this problem will support the efficient use of chemical fertilizers for farmers in age 4.0, the automatic self-cut fertilizers are featured in the method of use in which users can set the filling quantity and select the data. From more than 20 types of plants required for the calculation of the appropriate fertilizer proportion Easily and quickly.

Producer: Aj.Acha Watakit / Mr. Phatchai Taweepee

2.3 Related Technology

2.3.1 Arduino Mega 2560 [1]

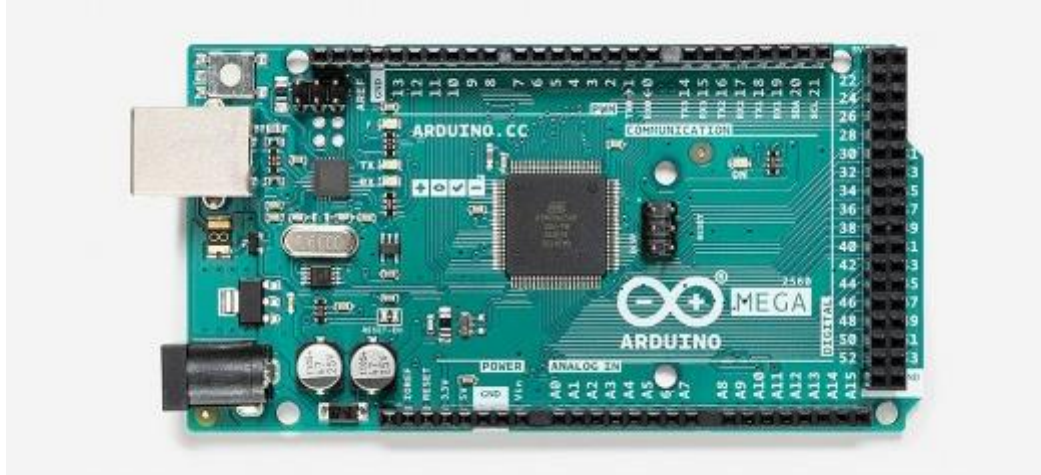


Figure 2.4 Arduino Mega 2560 REV 3

The Arduino Mega 2560 is the 8-bit board with 54 digital pins, 16 analog inputs, and 4 serial ports. It is a microcontroller board based on the ATmega2560. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analog inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Mega 2560 board is compatible with most shields designed for the Uno and the former boards Duemilanove or Diecimila.

Features & Parametrics

Features:

1. High Endurance Non-volatile Memory Segments
 - Write/Erase Cycles: 10,000 Flash
2. Atmel QTouch library support
3. JTAG (IEEE std. 1149.1 compliant) interface
4. Peripheral Features
 - Real-time Counter with Separate Oscillator
 - Programmable Watchdog Timer with Separate On-chip Oscillator
 - On-chip Analog Comparator
 - Interrupt and Wake-up on Pin Change
5. Other special features
 - Power-on Reset and Programmable Brown-out Detection
 - Internal Calibrated Oscillator
 - External and Internal Interrupt Sources
 - Six Sleep Modes: Idle, ADC Noise Reduction, Power-save, Power-down, Standby, and Extended Standby

Parametrics:

Name	Value
Program Memory Type	Flash
Program Memory Size (KB)	256
CPU Speed (MIPS/DMIPS)	16
SRAM (KB)	8,192
Data EEPROM/HEF (bytes)	4,096
Digital Communication Peripherals	4-UART, 5-SPI, 1-I2C
Capture/Compare/PWM Peripherals	4 Input Capture, 4 CCP, 16PWM
Timers	2 x 8-bit, 4 x 16-bit
Number of Comparators	1
Temperature Range (°C)	-40 to 85
Operating Voltage Range (V)	1.8 to 5.5
Pin Count	100

Table 2.1 Arduino Mega 2560 Parametrics**Advantages:**

- Low power consumption with fast start-up.
- Easier to use, with 8-bit microcontroller being less complex than 32/64-bit versions.
- QTouch Suite allows for ease of exploring, developing and debugging own touch applications.
- Patented Adjacent Key Suppression technology allows for unambiguous detection of key events.

Disadvantages:

- Limited amount of flash memory write cycles restricts the number of times images can be flashed when programmed to pc.
- Naturally lacks incremental performance compared to higher bit microcontrollers.

CHAPTER 3

METHODOLOGY

3.1 System overview

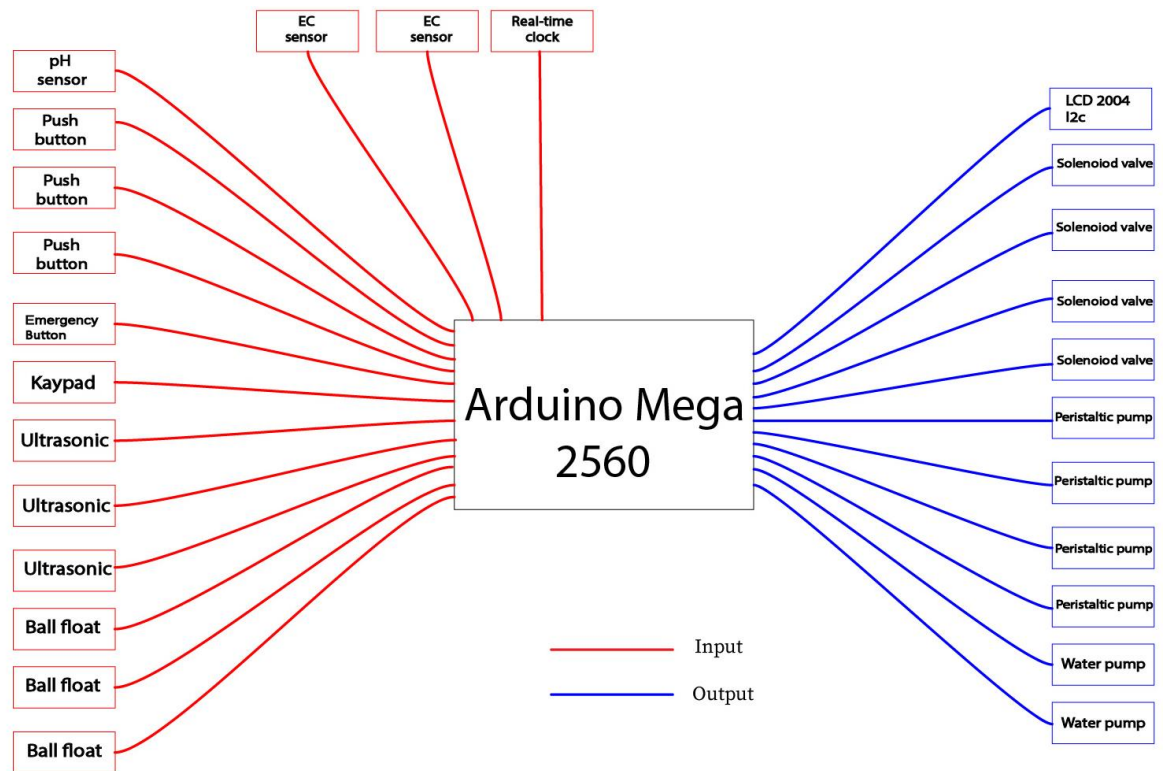


Figure 3.1 Intelligent watering system overview

3.1.1 Inputs: There are 15 inputs in this system as followings:

3.1.1.1 Analog pH sensor X 1



Figure 3.2 Analog pH sensor PE HOSE

3.1.1.2 EC sensor X 2



Figure 3.3 EC sensor - Grove V1.0

3.1.1.3 Push button X 3



Figure 3.4 Push button 1NO 1NC

3.1.1.4 Emergency stop button X 1



Figure 3.5 Emergency stop button

3.1.1.5 Keypad X 1



Figure 3.6 Emergency stop button

3.1.1.6 Ultrasonic Sensor X 3



Figure 3.7 HC-SR04 Ultrasonic sensor

3.1.1.7 Ball float X 3



Figure 3.8 10 cm ball float

3.1.1.8 Real-time clock X 1



Figure 3.9 DS3231 RTC

3.1.2 Processor: Arduino Mega 2560 Rev 3 microcontroller



Figure 3.10 Arduino Mega 2560 Rev 3

3.1.3 Outputs: There are 14 inputs in this system as followings:

3.1.3.1 Solenoid valve X 4



Figure 3.11 24VDC ½ solenoid valve

3.1.3.2 Peristaltic pump X4



Figure 3.12 24VDC Peristaltic pump

3.1.3.3 Water pump X2



Figure 3.13 0.5HP water pump

3.1.3.4 LCD 2004 I2C X 1



Figure 3.14 LCD 2004 I2C

3.2 System design

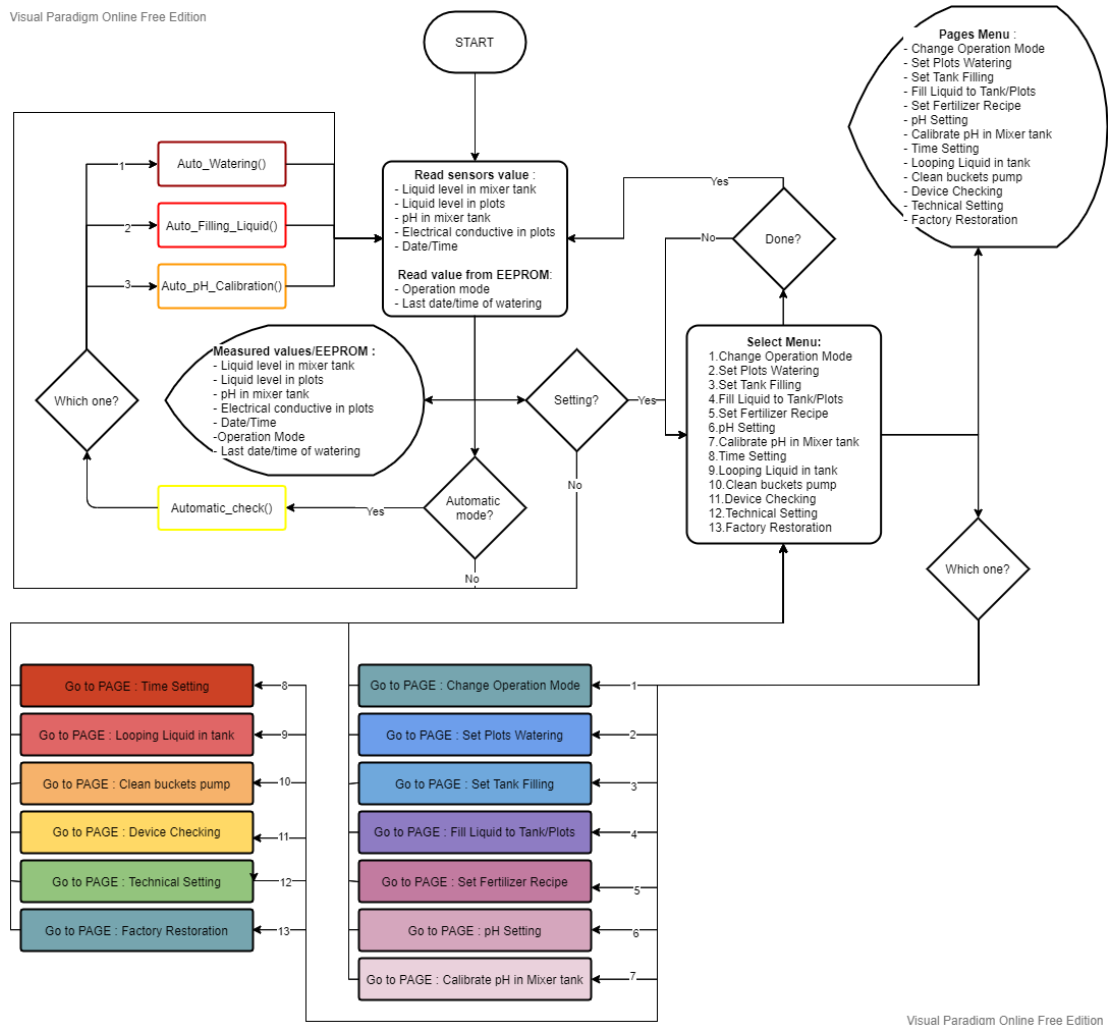


Figure 3.15 System flowchart that running continuously all the time

3.3 System Function

3.3.1 Changing operation mode

This function allow user to adjust operation mode as following:

- **Auto:** System will execute all main function task (Watering, pH calibration, Filling liquid into mixer tank, Maintain the liquid level in Mixer tank and plots to be on the criteria)
- **Manual:** User must command to the machine manually.
- **Hybrid:** System will execute some task automatically and other will operate by user command.

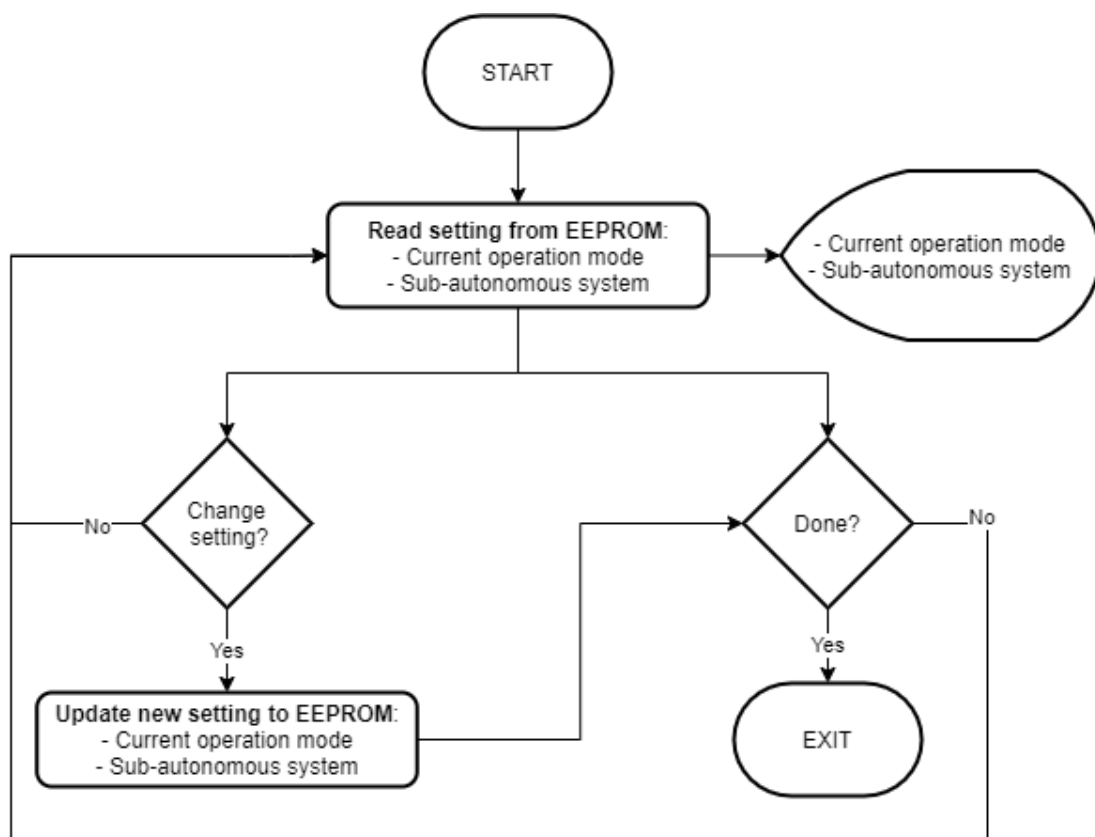


Figure 3.16 Systematic flowchart of Changing operation mode function

3.3.2 Setting plots watering

This function allow user to adjust setting of plots watering as following:

- **Water with:** Allow user to choose liquids to water into each plot (Fertilizer or Water)
- **Plot's liquid level:** User can adjust liquid level maintaining in each plot

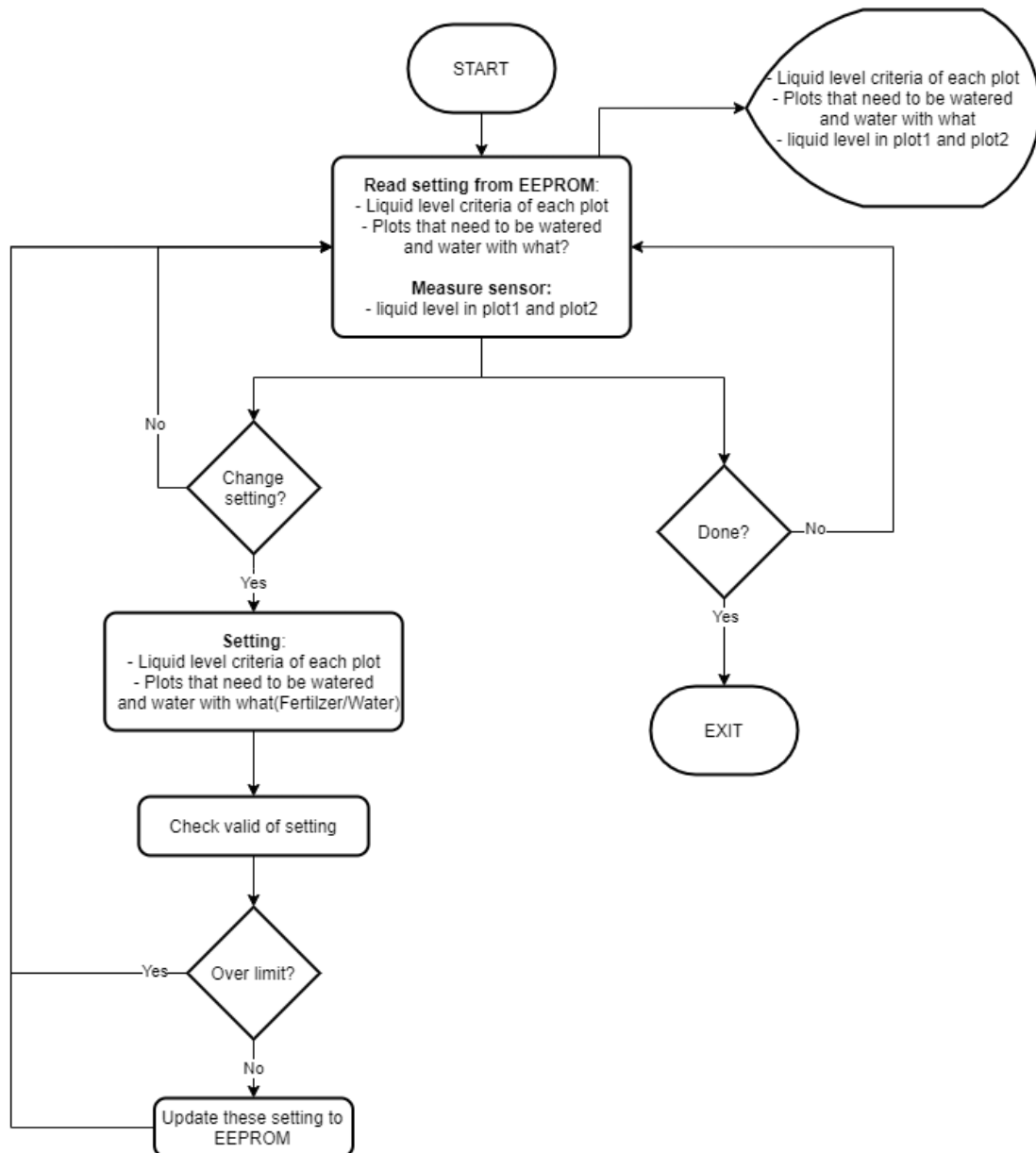


Figure 3.17 Systematic flowchart of Setting plot watering function

3.3.3 Setting mixer tank filling

This function allow user to adjust setting of mixer tank filling as following:

- **Fill with:** Allow user to choose liquids to fill into mixer tank (Fertilizer or Water)
- **Tank's liquid level:** User can adjust liquid level maintaining in mixer tank.

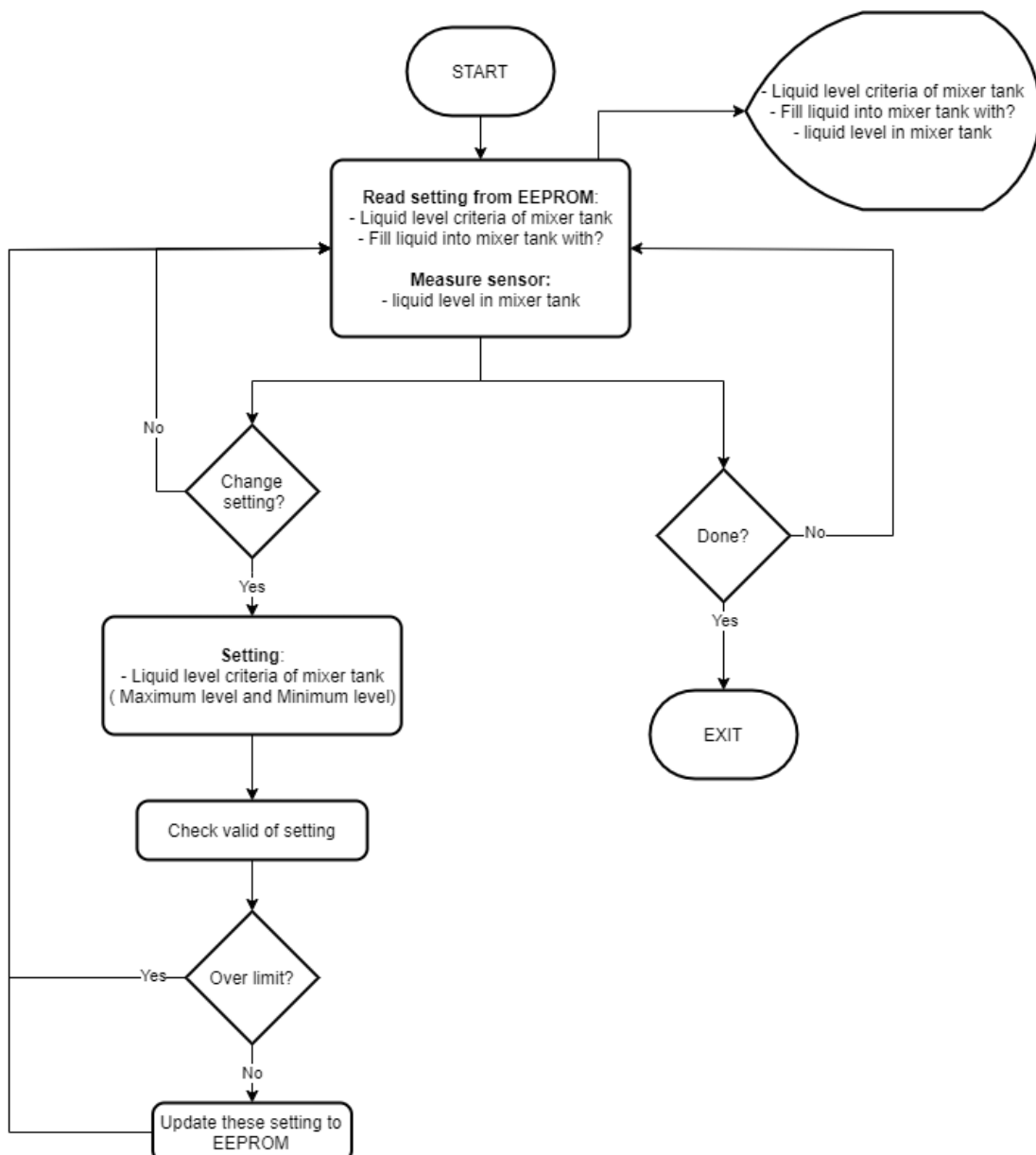


Figure 3.18 Systematic flowchart of Setting mixer tank filling function

3.3.4 Fill liquid into mixer tank or water liquid into plots

This function allow user to command the machine to fill liquid into mixer tank or water liquid into plots as following:

- **Select to Tank/Plots:** Allow user to select mixer tank or plots to fill the liquid.
- **Fill with:** Allow user to choose liquids to fill into mixer tank (Fertilizer or Water)

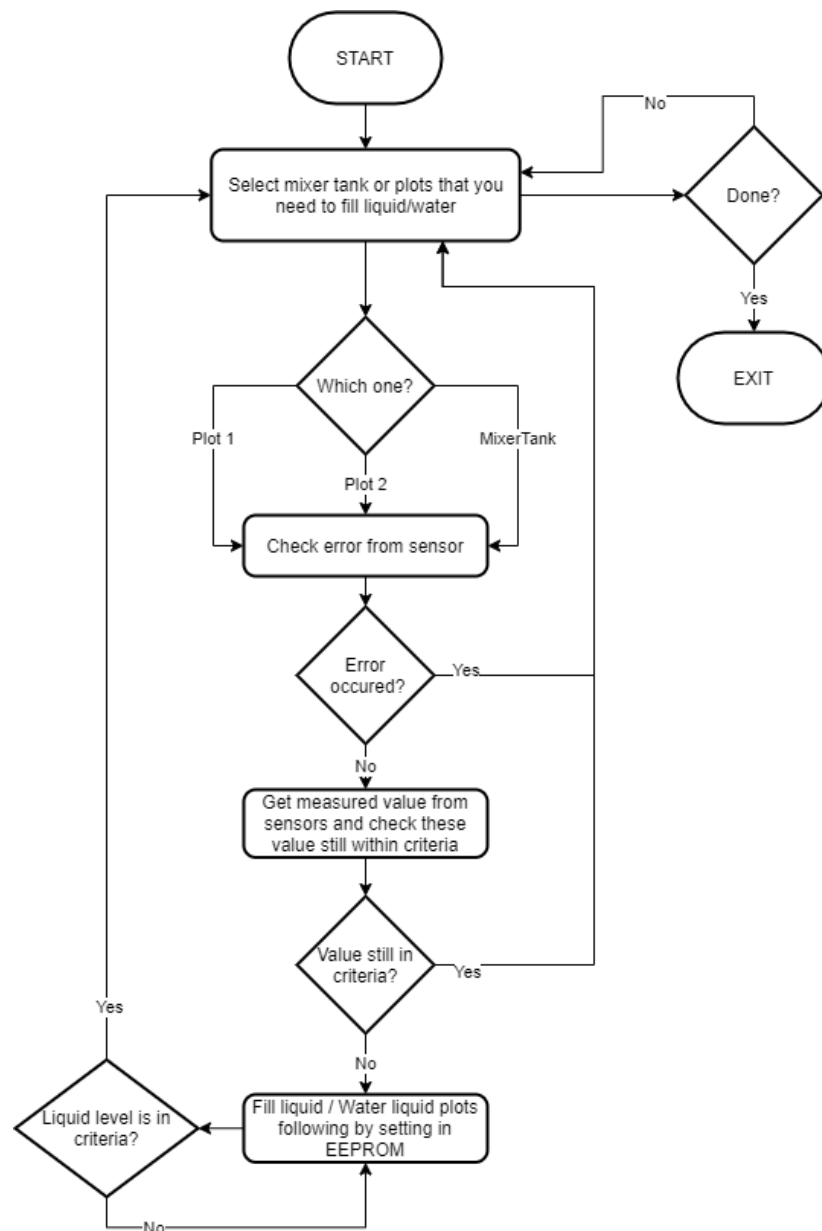


Figure 3.19 Systematic flowchart of filling liquid into mixer tank or watering liquid into plots function

3.3.5 Custom and selection of fertilizer recipe

This function allow user to select recipe of fertilizer in recipe list and you also can create your own recipe by functions as following:

- **Select recipe:** Allows user to select both built-in recipe and custom recipe.
- **Recipe creation:** Allow user to create their own fertilizer recipe.

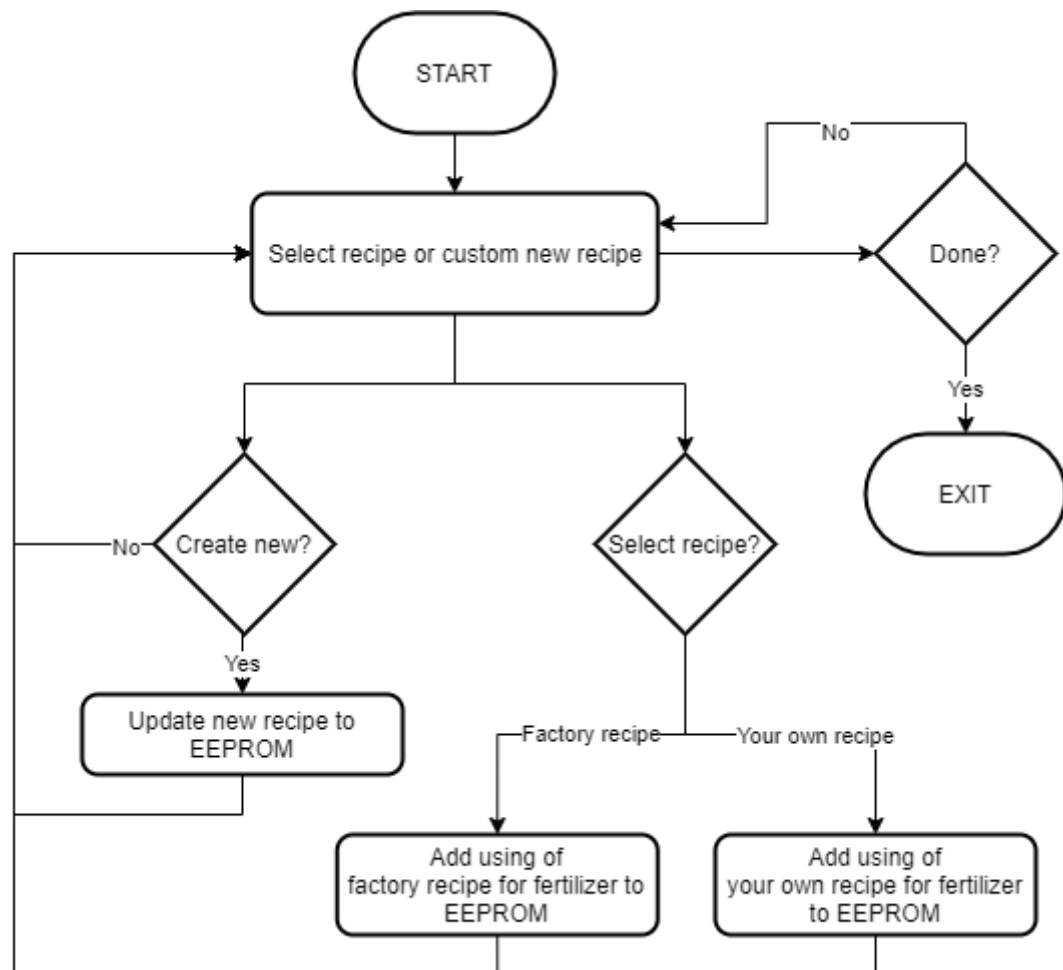


Figure 3.20 Systematic flowchart of custom and selection of fertilizer recipe function

3.3.6 pH value settings

This function allow user to settings all things about the pH value in the mixer tank which have following:

- **pH value criteria:** Allows user to define the number of pH value that will be the criteria of pH calibration. Which the machine must hold that pH level to be in criteria.
- **Differential:** Allow user to scope the range of criteria.

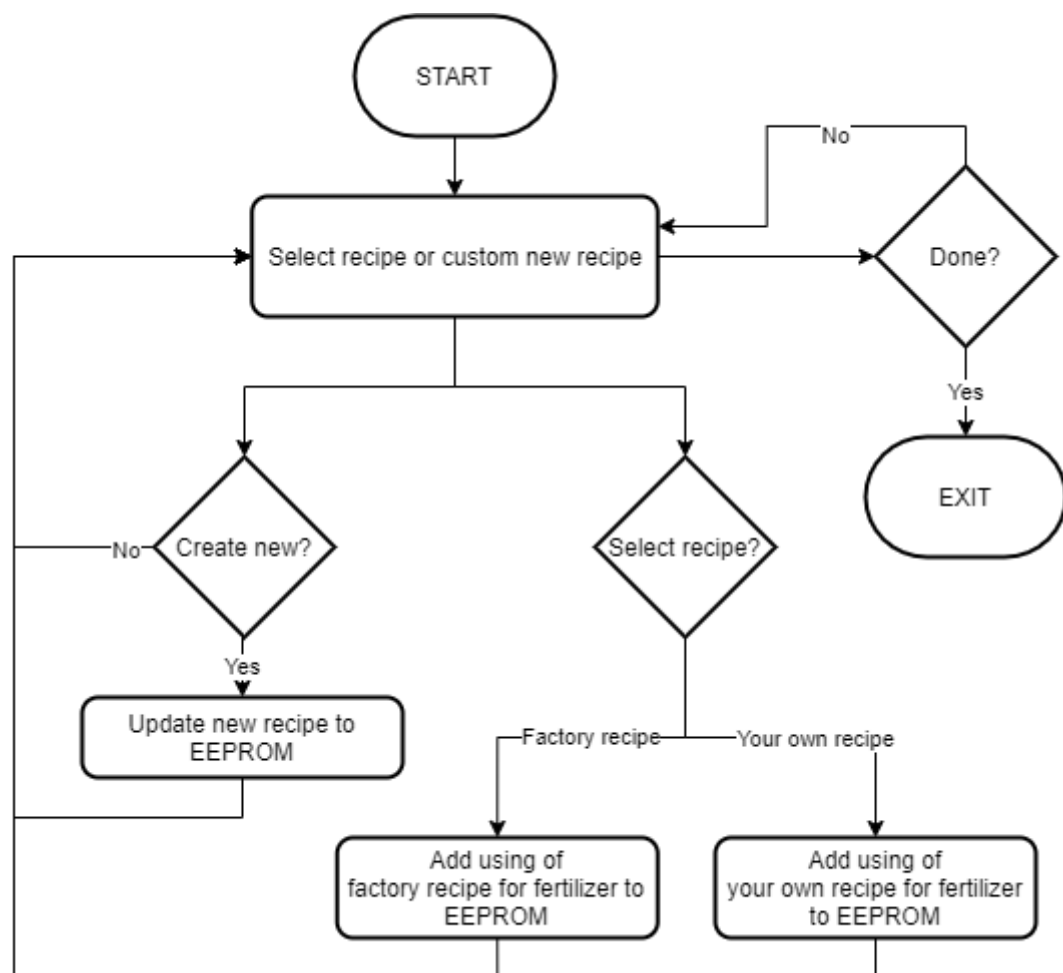


Figure 3.21 Systematic flowchart of setting pH value function

3.3.7 pH calibration

This function allow user to turn-on pH calibration function for calibrate pH value in the mixer tank to be in criteria from pH settings with conditions as following:

- **If pH value in tank + Differential < Criteria:** The machine will take the bases buffer into the mixer tank (In this case, the buffer mentioned before is tap water because the status of tap water is quite bases)
- **If pH value in tank - Differential > Criteria:** The machine will take the Acid buffer into the mixer tank.

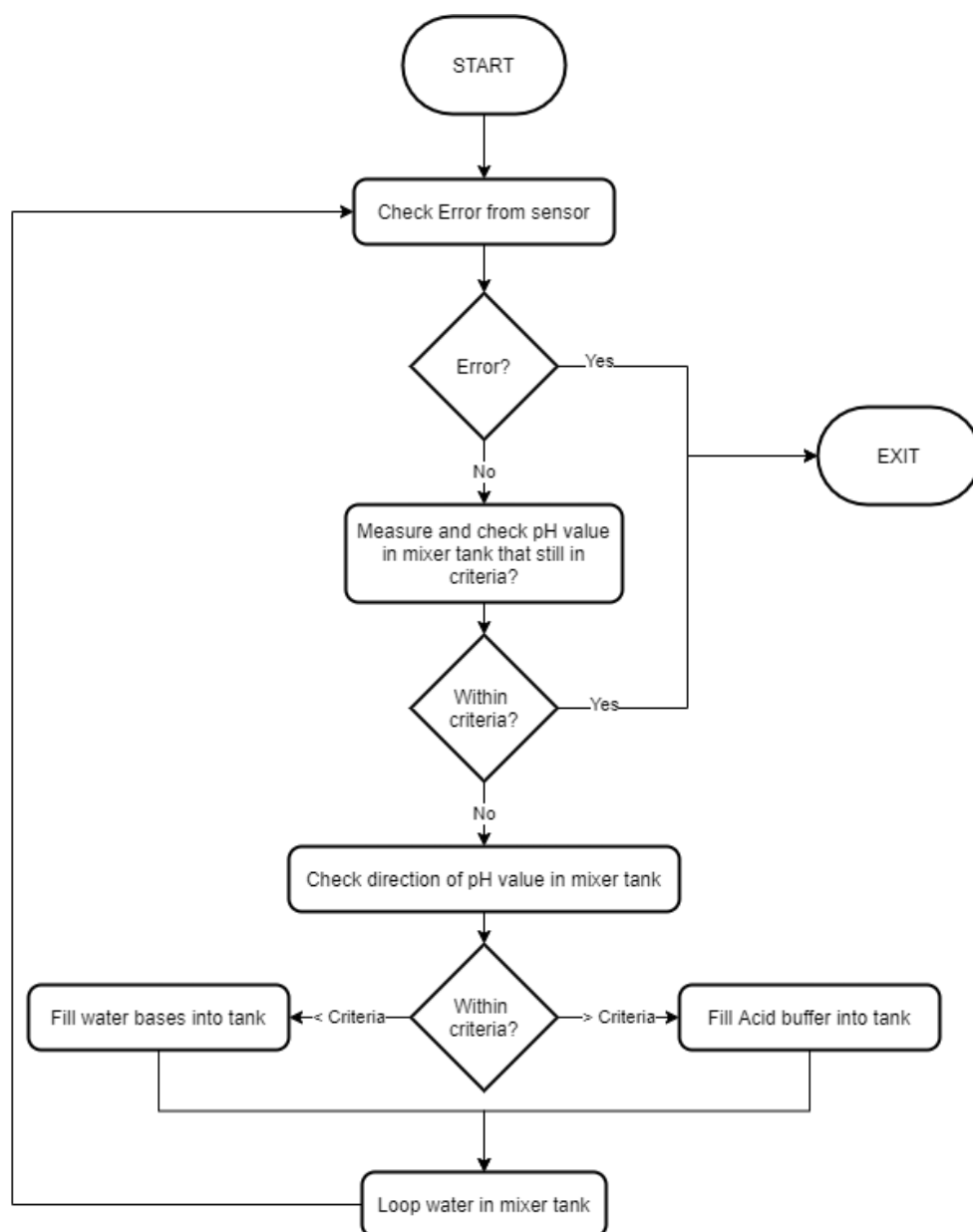


Figure 3.22 Systematic flowchart of setting pH value function

3.3.8 EC settings

This function allow user to settings all things about the EC value in each plot which have following:

- **Offset value:** Allows user to define the offset value for calibrate the analog data from EC sensor to be in standard.
- **Offset-Sign:** Define the sign of that offset value (Positive or Negative).

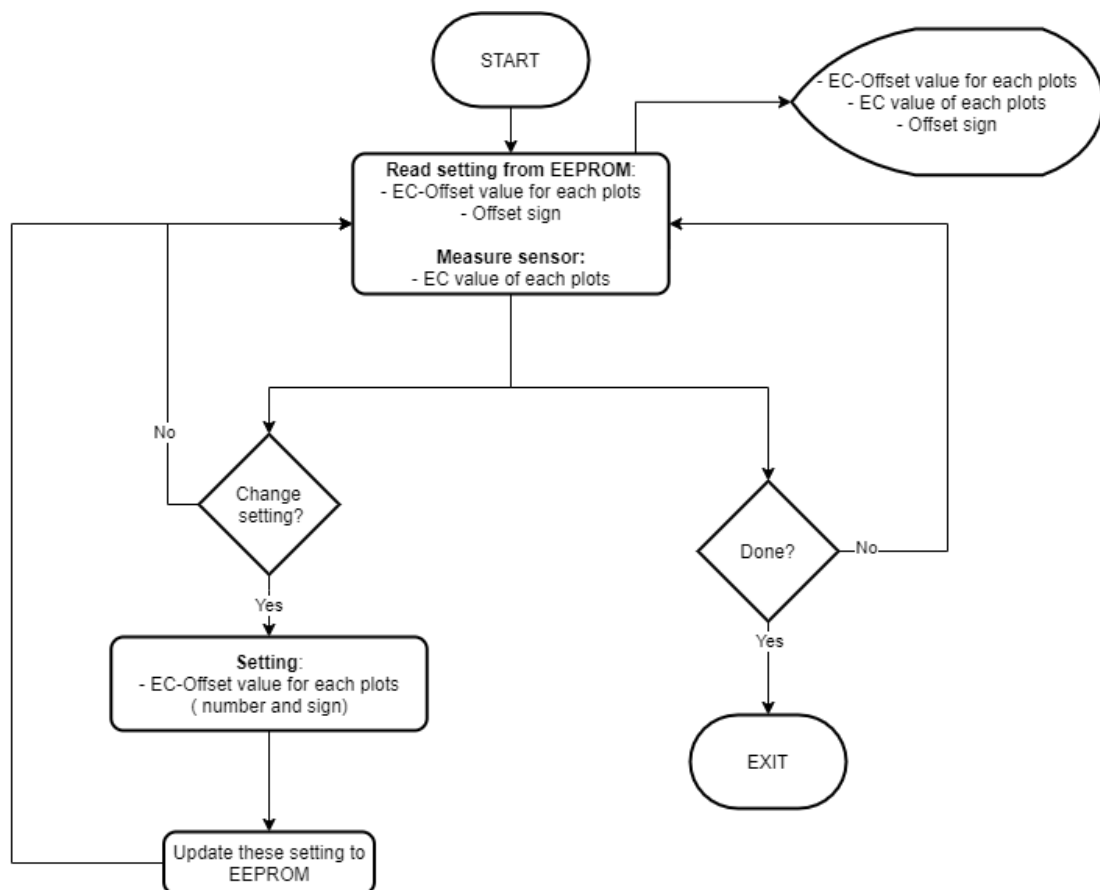


Figure 3.23 Systematic flowchart of setting EC value function

3.3.9 Looping liquid in the mixer tank

This function allow user to turn-on liquid looping function for stir the fertilizer in mixer tank to prevent the precipitation of the chemical fertilizer.

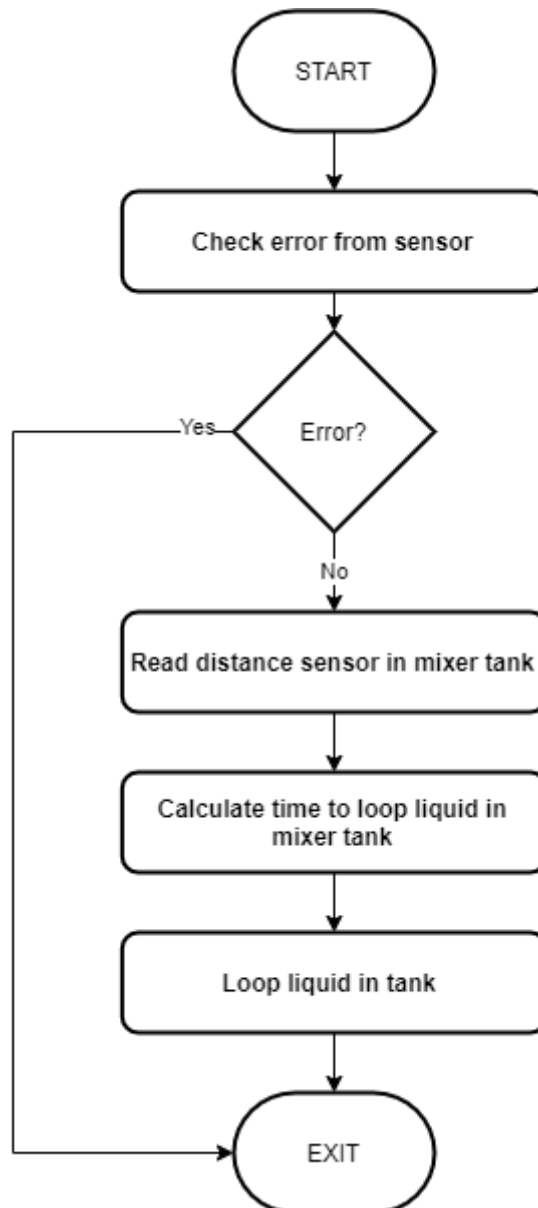


Figure 3.24 Systematic flowchart of looping liquid in the mixer tank function

3.3.10 Buffer pump cleaning

This function allow user to command the machine to expel liquid in the hose of buffer pump.

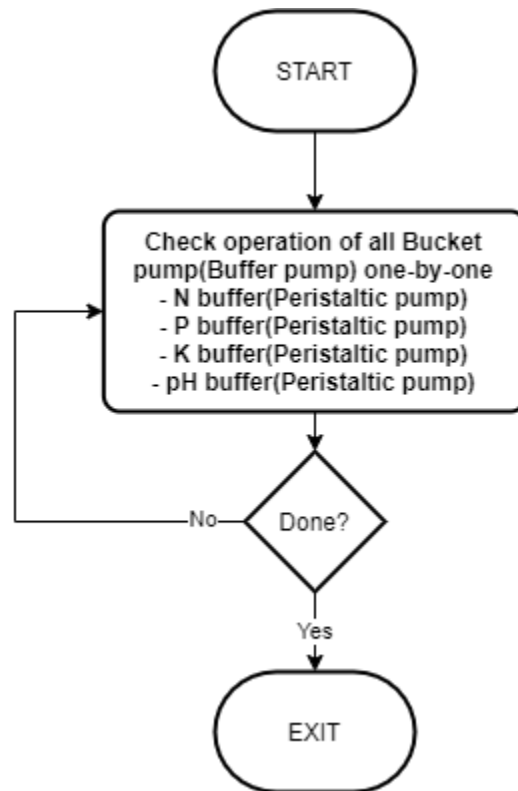


Figure 3.25 Systematic flowchart of expel liquid in the hose of buffer pump function

3.3.11 Testing essential devices

This function allow user to check all essential devices that it still working or not.

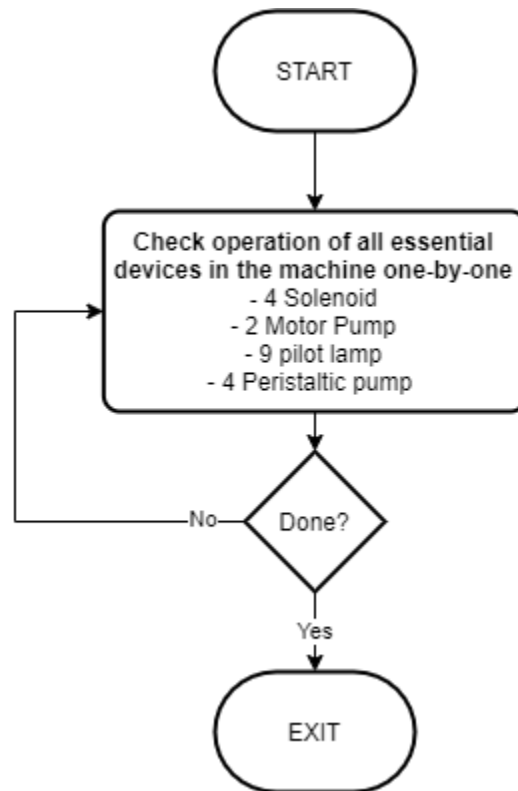


Figure 3.26 Systematic flowchart of testing essential device's function

3.3.12 Time setting

This function allow user to set date and time of the machine.

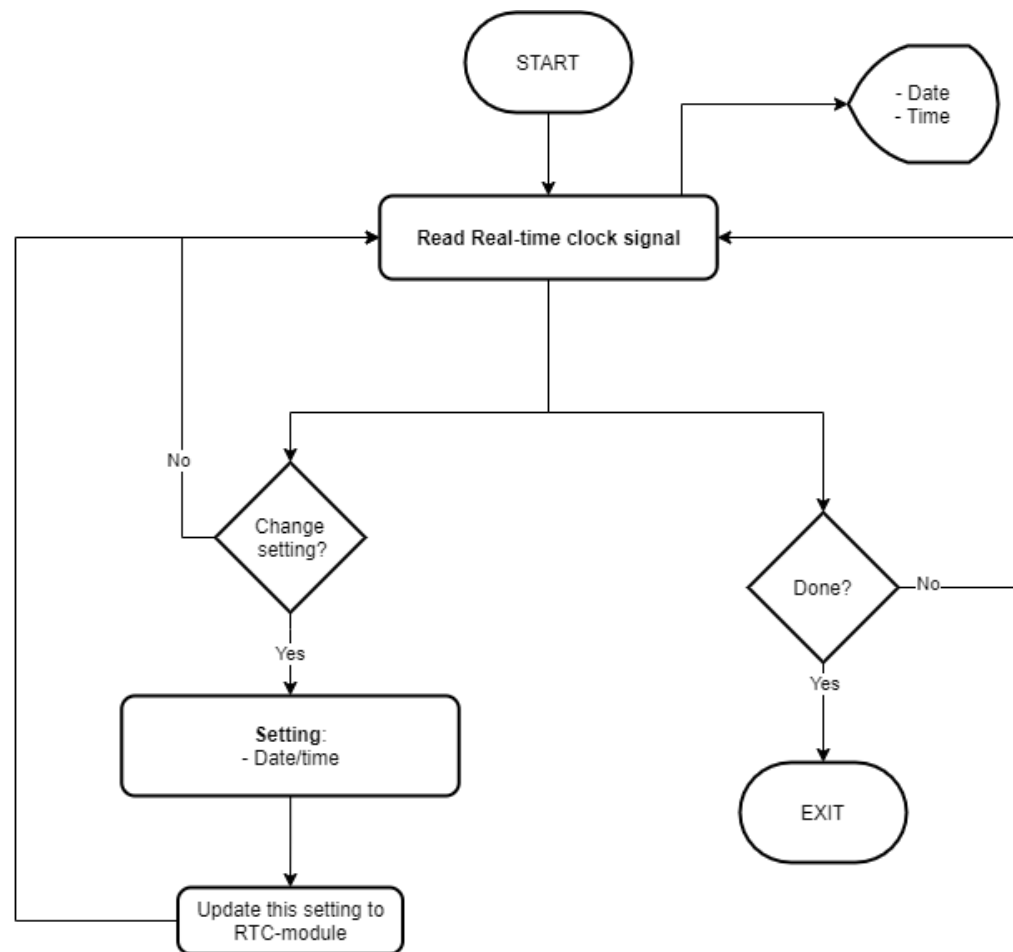


Figure 3.27 Systematic flowchart of time setting function

3.3.13 Technical settings

This function allow user to set the important constant of the machine that used for calculation in some sub-system.

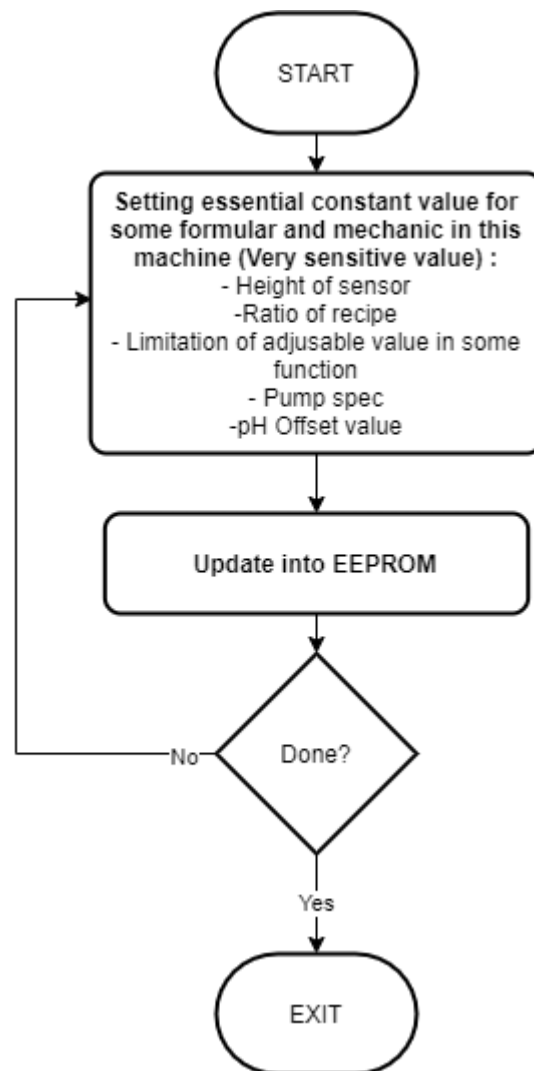


Figure 3.28 Systematic flowchart of technical settings function

3.3.14 Factory restoration

This function allow user to restore all setting and constant in the machine to be factory settings.

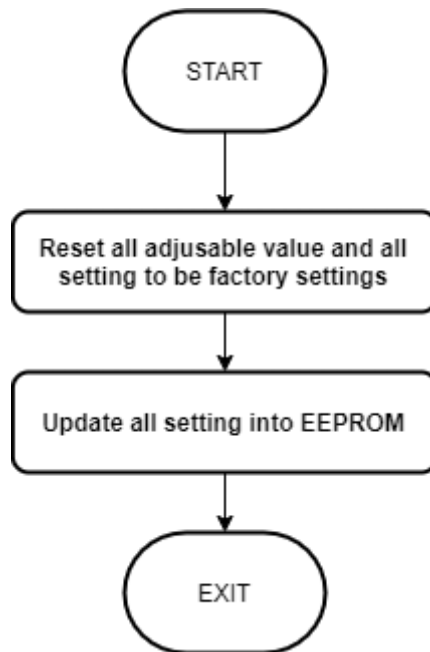


Figure 3.29 Systematic flowchart of factory restoration function

3.4 System schematic

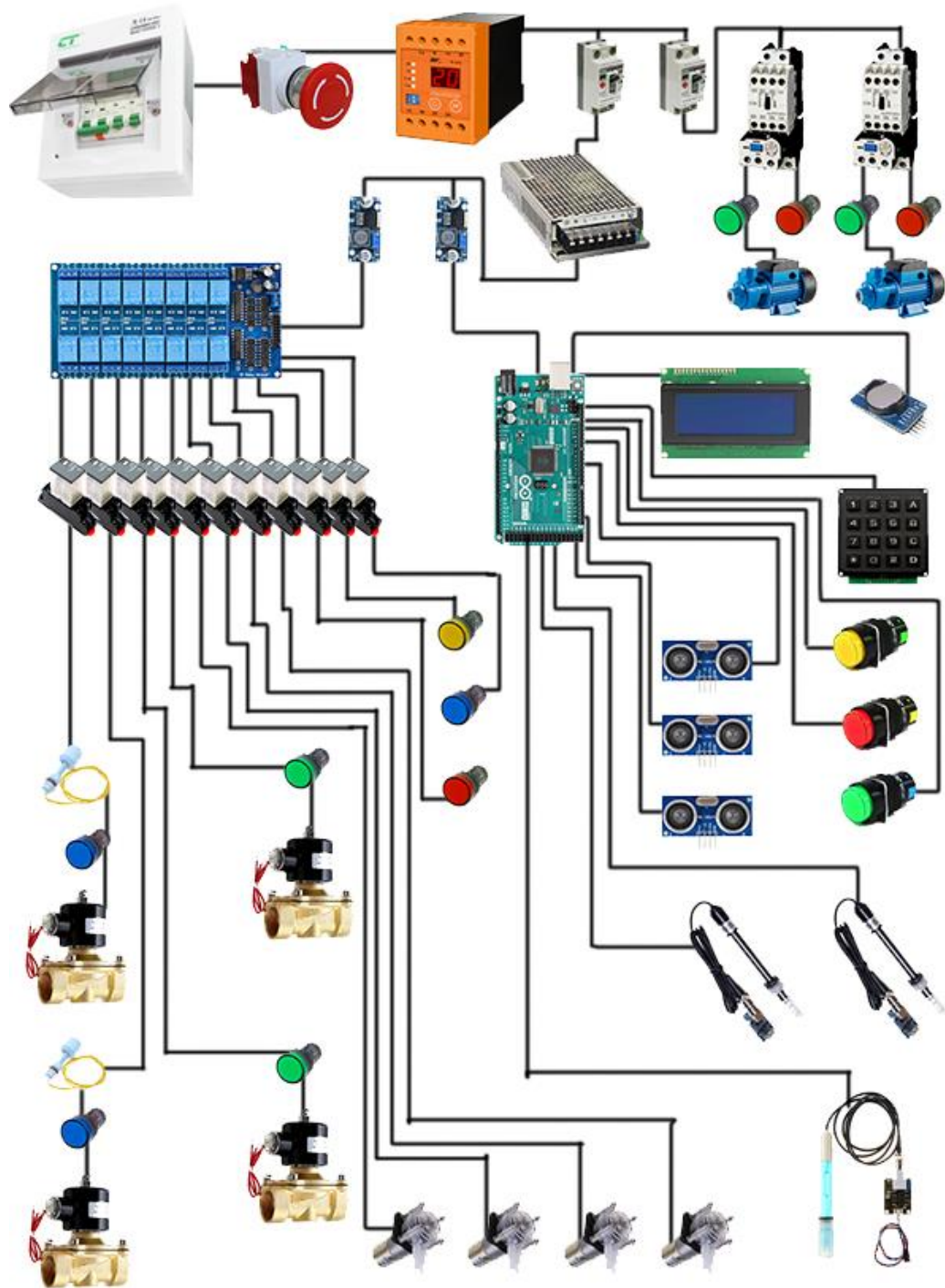


Figure 3.30 System schematic of intelligent watering system

3.5 User Interface design

3.5.1 Main page

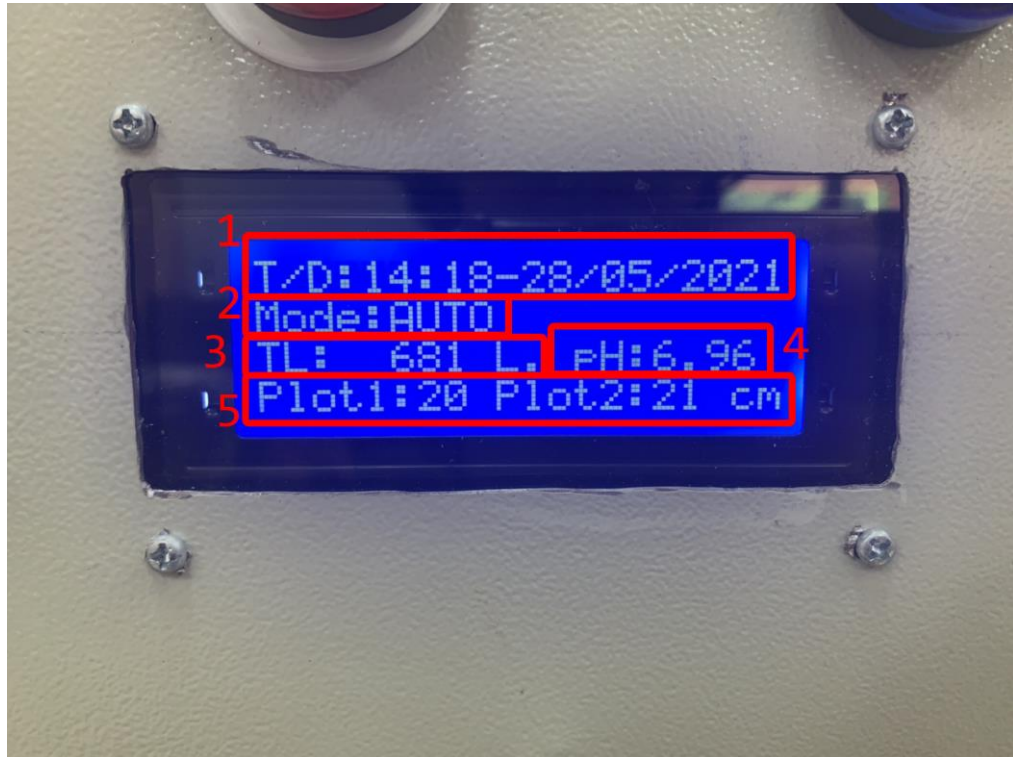


Figure 3.31 Components of main page

Description: This screen is the main screen that show value from sensor measurements and allow the machine to do automation task.

Components:

1. Real-time clock of the system that shows date and time (HH:MM – dd/mm/yyyy)
2. It is an operation mode that show the current mode of system.
3. This part is showing of level of liquid in the mixer tank and update every 10 second.
4. pH value which was measured by sensor and update every 10 second.
5. This part is showing of liquid level in each plot by centimeter unit and update every 10 second.

3.5.2 Page Menu of settings

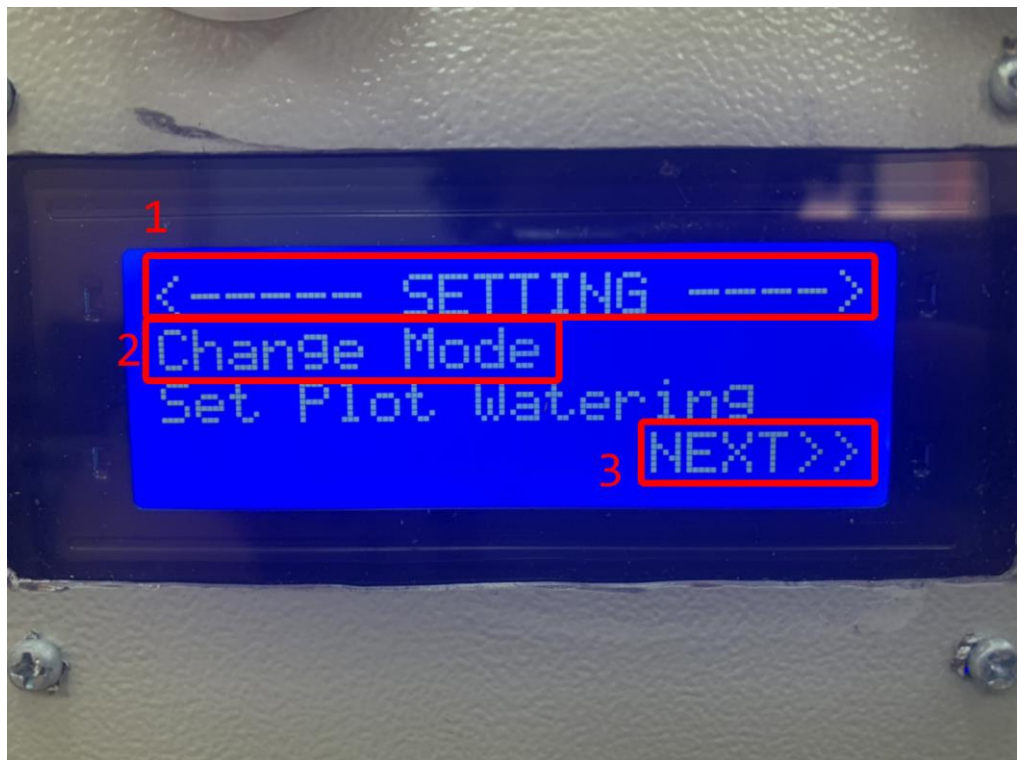


Figure 3.32 Page Menu of settings

Description: This screen will be opening when user need to adjust somethings in system.

Components:

1. Label to show that you are in the menu of setting.
2. Category topic for setting.
3. Going to next page.

3.5.3 Task progress pattern 1

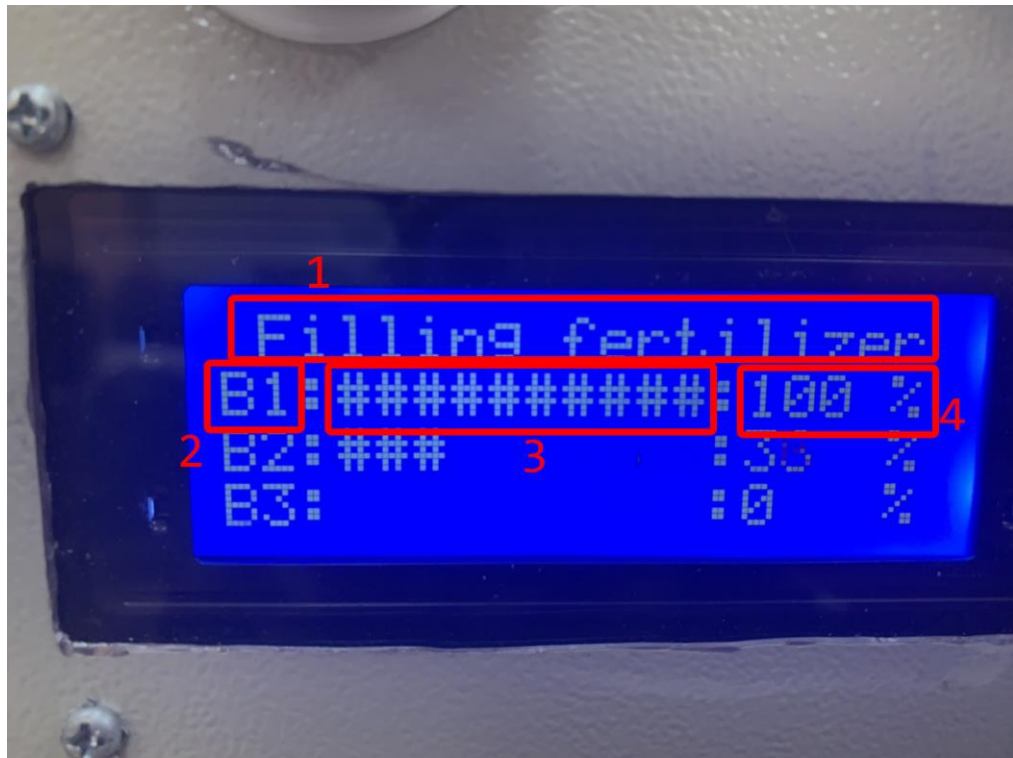


Figure 3.33 Screen of task progress pattern 1

Description: This pattern is uses in progress of task with a fixed time schedule.

Components:

1. Label to show what task that the machine is working on.
2. Label of what thing that machine is working on (In this case "B1" is mean Barrel 1)
3. Symbol that is representing progress of that task.
4. Percentage of task progress.

3.5.4 Task progress pattern 2

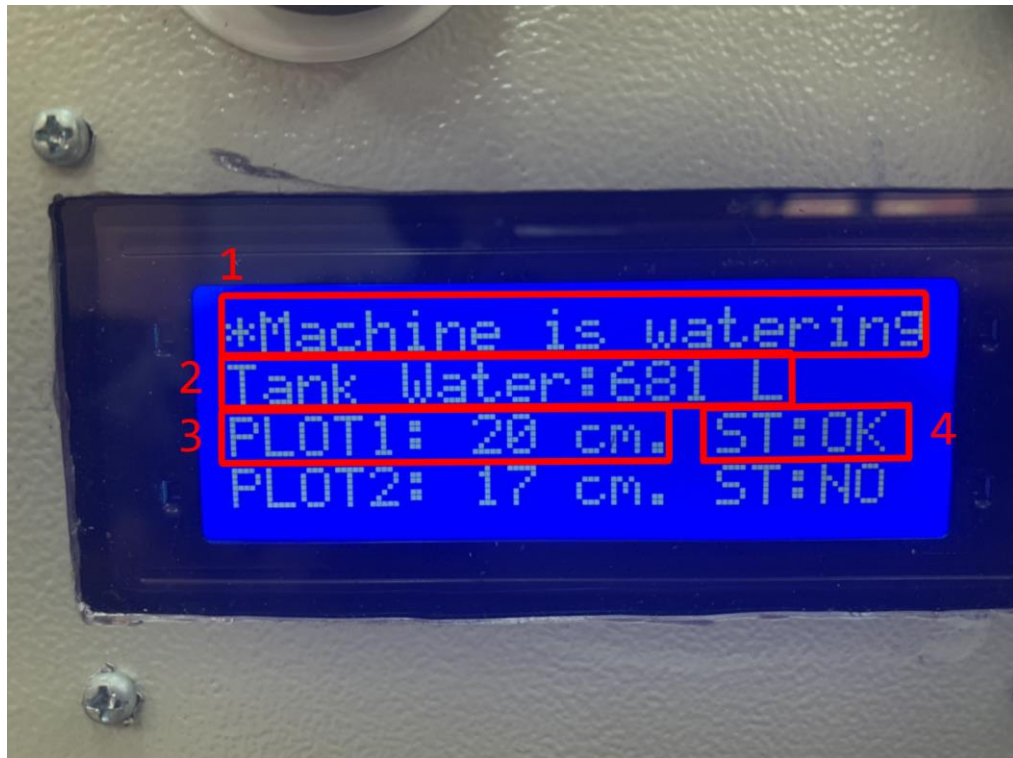


Figure 3.34 Screen of task progress pattern 2

Description: This pattern is used in progress of plot watering only.

Components:

1. Label to show what task that the machine is working on.
2. Level of liquid in mixer tank with Litre unit.
3. Level of liquid in each plot with Centimeter unit.
4. Status of the plot to tell user that level of liquid in that plot is in the criteria.

3.5.5 Task progress pattern 3

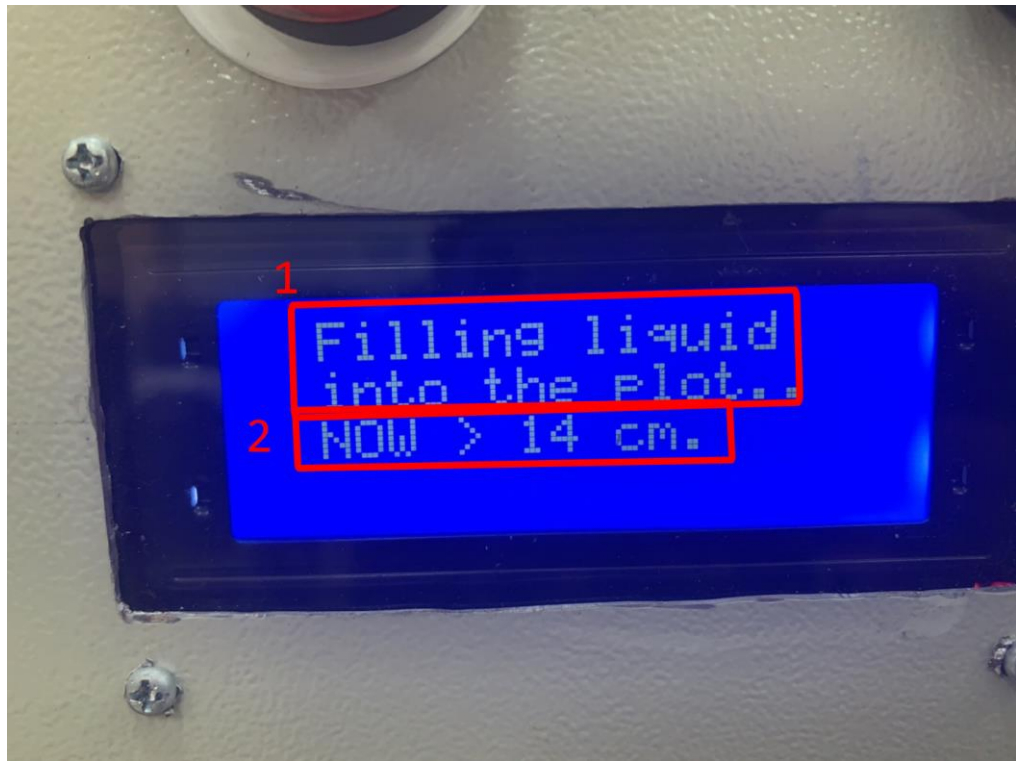


Figure 3.35 Screen of task progress pattern 3

Description: This pattern is used in the progress of filling liquid into the tank only.

Components:

1. Label to show what task that the machine is working on.
2. Level of liquid in mixer tank with Litre unit.

3.5.6 Blinking text

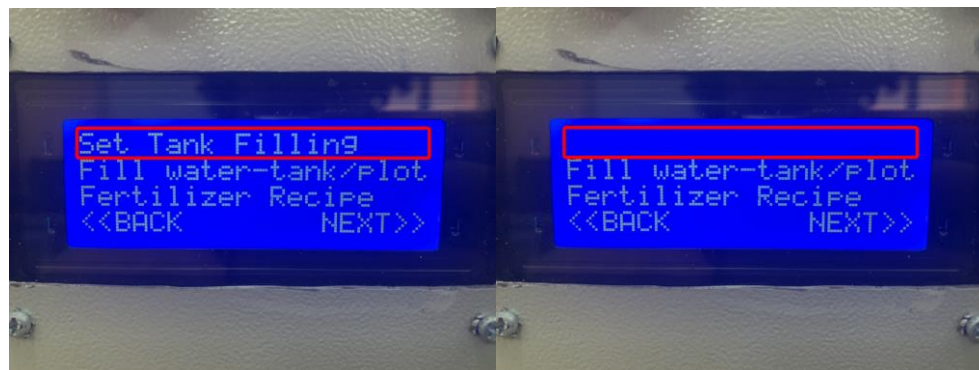


Figure 3.36 Blinking text

Description: Blinking text is indicating of current line.

3.5.7 Notification screen



Figure 3.37 Notification screen that appear when the task is done

Description: Notification screen is the screen that will be appearing after task done.

3.6 Manufacturing Process

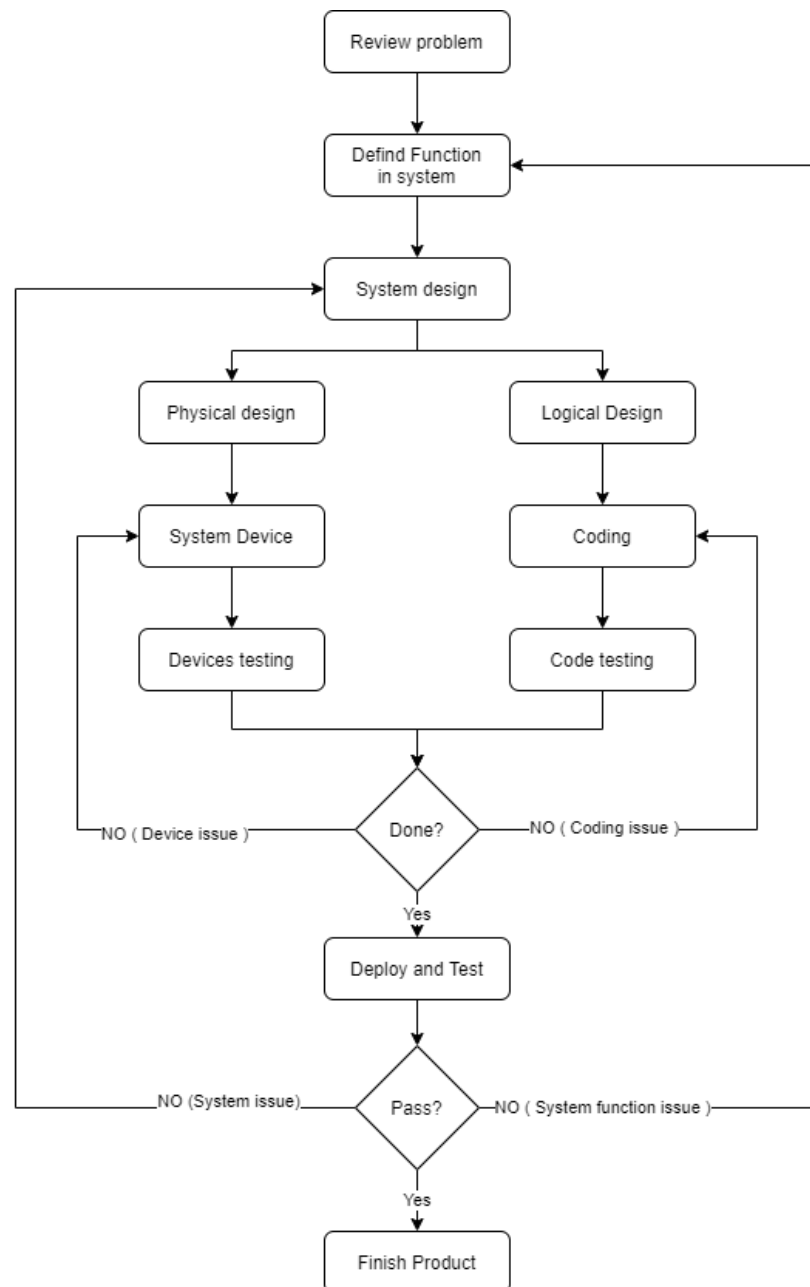


Figure 3.38 Manufacturing flowchart from beginning to finished product

3.7 Formulations

3.7.1 Liquid level in plot

Required variables:

- Height of sensor (Centimeter) (by user)
- Distance of sensor to liquid level (Centimeter) (by sensor)

Formular:

Give x = Liquid level in plot (Centimeter)

$$F(x) = \text{Height of sensor} - \text{Distance of sensor to liquid level}$$

3.7.2 Liquid level in mixer tank

Required variables:

- Height of sensor (Centimeter) (by user)
- Distance of sensor to liquid level (Centimeter) (by sensor)
- Height of mixer tank (Centimeter) (by user)
- Mixer tank capacity (Centimeter) (by user)

Formular:

Give x = Liquid level in mixer tank (Littre)

$$F(x) = (\text{Height of sensor} - \text{Distance of sensor to liquid level}) \times \left(\frac{\text{Height of mixer tank}}{\text{Mixer tank capacity}} \right)$$

3.7.3 Recipe Ratio

Required variables:

- New Fertilizer Recipe Ratio (Littre) (by user)

Formular:

Give x = Recipe ratio (Littre)

$$F(x) = \frac{\text{New Fertilizer Recipe Ratio}}{1000}$$

3.7.4 Current recipe ratio

Required variables:

- Criteria Maximum level in mixer tank (Litre) (by user)
- Liquid level in mixer tank (Litre) (Calculated in machine)

Formular:

Give x = Current recipe ratio (Litre)

$$F(x) = \frac{\text{Liquid level in mixer tank}}{\text{Criteria Maximum level in mixer tank}}$$

3.7.5 Actual recipe ratio

Required variables:

- New Fertilizer Recipe Ratio (Litre) (by user)
- Criteria Maximum level in mixer tank (Litre) (by user)

Formular:

Give x = Actual Recipe ratio (Litre)

$$F(x) = \frac{\text{Criteria Maximum level in mixer tank}}{\text{New Fertilizer Recipe Ratio}}$$

3.7.6 Suction of substance

Required variables:

- *Substance Volume_i* = Type of substance in each barrel (Milliliter) (by user / built-in)
- Recipe Ratio (Litre) (Calculated in machine)
- Actual Recipe Ratio (Litre) (Calculated in machine)

Formular:

Give x = Amount of substance pumping (Milliliter)

$$F(x) = (\text{Recipe Ratio} \times \text{Substance Volume}_i) \times \text{Actual Recipe Ratio}$$

3.7.7 Actual pumping of substance

Required variables:

- Pumping of substance (Milliliter) (Calculated in machine)
- Current recipe ratio (Litre) (Calculated in machine)

Formular:

Give x = Actually amount of substance pumping (Milliliter)

$$F(x) = \text{pumping of substance} \times (1 - \text{Current recipe ratio})$$

3.7.8 Pumping interval time

Required variables:

- Actual pumping of substance (Milliliter) (Calculated in machine)
- Peristaltic pump capacity (Milliliter / Minute) (by user)

Formular:

Give x = Time interval to pumping fertilizer into mixer tank (Millisecond)

$$F(x) = \left(\frac{60,000}{\text{Peristaltic pump capacity}} \right) \times \text{Actual pumping of substance}$$

3.7.9 Stirring interval time

Required variables:

- Liquid level in mixer tank (Litre) (Calculated in machine)
- Motor pump capacity (Litre/Hour) (by user)

Formular:

Give x = Time interval to stir the liquid in mixer tank (Millisecond)

$$F(x) = \left(\frac{\text{Liquid level in mixer tank}}{\text{Motor pump capacity}} \right) \times 60,000$$

CHAPTER 4

EXPERIMENT AND RESULTS

4.1 Experiments

4.1.1 Continuous working

Continuous working is used for test the processing performance of processor or controller.

Method: Command the machine to do the all-possible tasks continuously for 1 hour.

4.1.2 Watchdog testing

This experiment is used for freezing of process testing of the processor or controller when we had turned it on for many days.

Method: Turn on machine for 7 days and count the amount of watchdog resetting.

4.1.3 Power and safety testing

Power and safety testing is used for test the source power sufficient for the entire system and safety while you are operating with machine.

Method: Write test case to turn on all devices in the same time and check all risk point in the machine that probably to be electric leakage.

4.1.4 Essential devices testing

Essential devices testing is used for test the efficiency of all essential devices.

Method: Write test case to command that devices then compare the result of that task with the expect result and capacity of that devices(ideal).

4.1.5 Function testing

Function testing is used for test the operation of all functions and tasks that it working correctly.

Method: Command the machine to do all-possible tasks and debug.

4.1.6 User interface testing

User interface testing is used for test the interface that it can act and display correctly.

Method: Try to Get in all interfaces in the machine and also check these interfaces can working related to current tasks.

4.1.7 Objectives review

Objective review is rechecking finished product can serve all objectives of project.

Method: Make a checklist of all objectives.

4.2 Results

4.2.1 Continuous working

Result: After start running test case to command the machine to execute tasks for 1 hour. The machine can finish all tasks without interrupt from processor sequence or load on devices.



Figure 4.1 Continuous operation of the machine for testing

4.2.2 Watchdog testing

Test schedule: Start turn it on at 10 May 2021 – 16 May 2021 (Check every day)

Result: After 7 days that we had turned it on, the machine still working continuous and can receive the user command then do the task correctly.



Figure 4.2 Machine after ran over 7 days and still working continuously

4.2.3 Power and safety testing

Result:

- Power: After turning on all devices at the same time, the current consumption in the part of controller circuits is about 5 Amp but the maximum capacity of power-supply is 10 Amp. Therefore, there is no problem if the machine has reach to possible consumption and have no spike or noise feedback to the source because this machine has installed surge suppressors devices to protect the power-supply from noise signal.



Figure 4.3 Machine with turning on all possible devices to operate in the same time

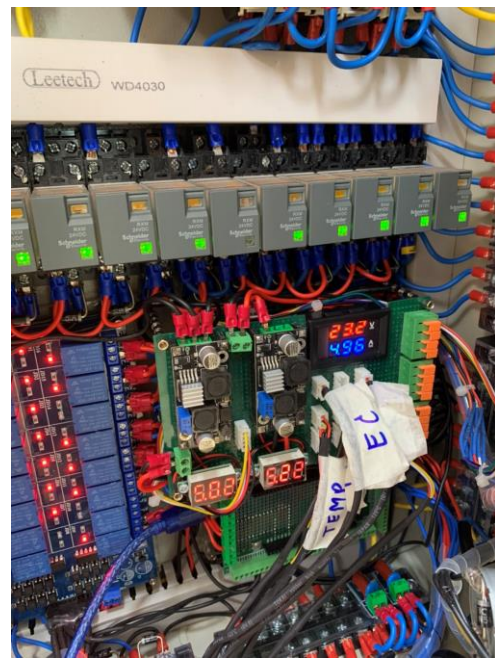


Figure 4.4 Total current consumption and voltage of the power supply that display on the digital meter

- Safety: There are some risk points from electric leakage such as screw head that has been protruding outside the container. But there is no electric flow through those points. All wires and connecting points have a casing and ground connection to prevent electric leakage either.



Figure 4.5 10A Circuit breaker, 1-Phase protection



Figure 4.6 Ground connection on the machine case



Figure 4.7 RCBO consumer unit for cut-off power line of machine when there is an electric leakage



Figure 4.8 Screw head that has been protruding outside the container

4.2.4 Essential devices testing

Result:

- Mechanic devices

1.Peristaltic pump: There are 4 peristaltic pumps in this machine for pumping the fertilizer substances to fill into the mixer tank and test pumping capacity of all peristaltic pumps. As the result, all peristaltic pump can pump liquid as specified capability at 500 ml/minute.



Figure 4.9 24VDC Peristaltic pump for pumping the fertilizer substance and pH substance into the mixer tank

2.Pilot lamp: There are 11 lamps in this machine that can light up with the operation of each device and turn off when those devices have stopped working without problem.



Figure 4.10 220 VAC Pilot lamp for operations notify

3.Pump motor: There are 2 pump motor in this machine. First one is the pump for tap water and another one for fertilizer. As the result, first motor and second one can operate correctly either.



Figure 4.11 Mitsubashi water pump

4.Solenoid valve: There are 4 solenoids valve in this machine for manage flow direction of liquid in pipe. As the result, those solenoids valve can made to order correctly.



Figure 4.12 24 VDC – for ½ Inches pipe solenoid valve

- Sensor

1. Tank and plots ultrasonic sensor: Both tank and plots sensor are based on the ultrasonic sensor to measure distance by using ultrasound. As the result, Ultrasonic can measure distance pretty well but still had a little noise signal.

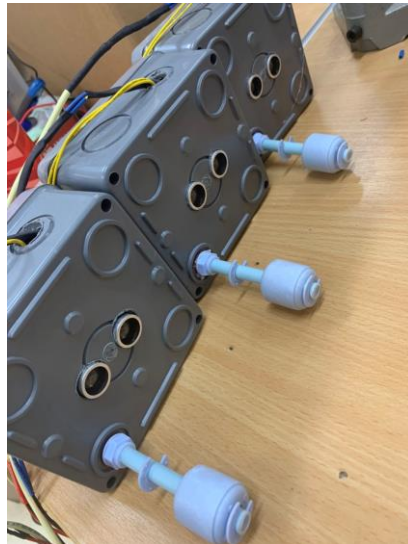


Figure 4.13 HC-SR04 Distance sensor with ball floats for mixer tank and plots



Figure 4.14 Application of plant pots with sensors

2. pH analog sensor: A pH analog sensor is quite problematic with measuring the constantly flowing water and generates a lot of noise in the large areas of water but the measurement in a specific substance that used to calibrate pH was very well.



Figure 4.15 Analog pH sensor PE HOSE for Arduino



Figure 4.16 Test and calibrate the pH sensor by immersion into tester buffer 4-7-10

Result of immersion test as followings:

Testing buffer 4-7-10	Sensor measurement	Standard deviation
4	-3.27 to -3.1	± 0.19
7	-0.21 to -0.13	± 0.12
10	2.44 to 3.05	± 0.25

Table 4.1 Before pH calibration

Testing buffer 4-7-10	Sensor measurement	Standard deviation
4	3.78 – 4.03	± 0.24
7	6.98 – 7.04	± 0.11
10	9.86 – 10.05	± 0.19

Table 4.2 After pH calibration

3. Electrical Conductive sensor: The EC sensor have a little bit noise signal with measuring the constantly flowing water and generates a lot of noise in the large areas of water likewise with pH sensor. As the result by dissolving 50 grams of the fertilizer in water and immersing the sensor in the cup, it turns out that the EC has increased clearly from 0-0.6 ms/cm before immersing the sensor tube upto 340.4-561.9 ms/cm after immersing the sensor tube (Still can't be calibrated because there is no test substance right now)



Figure 4.17 Electrical conductive sensor – Grove v1.0

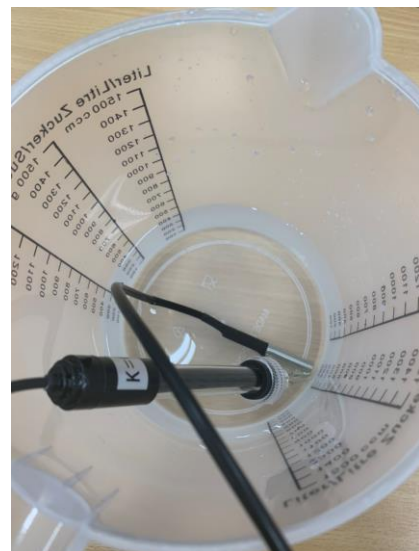


Figure 4.18 Testing the EC sensor by immersion into the water cub with temp sensor for calculation of EC value

4. Floating ball: This will prevent sensors from being damaged by malfunctions. This can happen when the amount of water in a mixer tank or in a plot exceeds the limit level by the ball floats. The working of the ball floats is not related to the process of the program at all. Rather, it will prevent current that will flow through to feed the pump motor.



Figure 4.19 Active state ball float



Figure 4.20 Cut-off state of ball float



Figure 4.21 Cut-off state is going to directly command to the relay

4.2.5 Function testing

Result: While running the program there was some little bug that occurred which has been corrected.

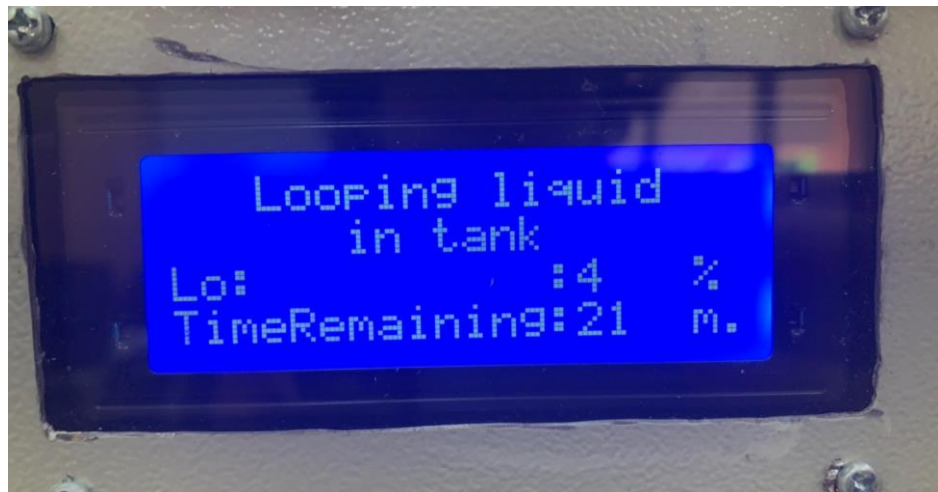


Figure 4.22 Example of error that had occurred in task process. In this case, percent of progress started with 4 percent instead of 0 or 1 percent

4.1.6 User interface testing

Result: We have used LCD screen as a user interface to display English characters as the main character for this machine. With the function of display the task that the machine is doing at that time and it is designed to be easy to use and there are few buttons to reduce user confusion.

Display of Menus and functions:



Figure 4.23 Main display



Figure 4.24 Another main display



Figure 4.25 Settings menu page 1

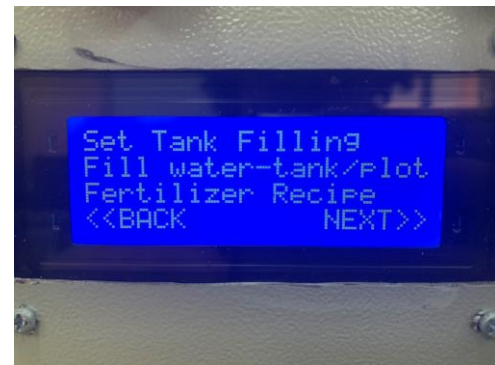


Figure 4.26 Settings menu page 2

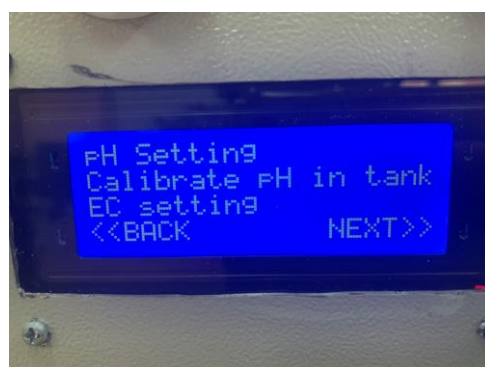


Figure 4.27 Settings menu page 3



Figure 4.28 Settings menu page 4



Figure 4.29 Settings menu page 5



Figure 4.30 Mode settings main page



Figure 4.31 Select Automatic function page

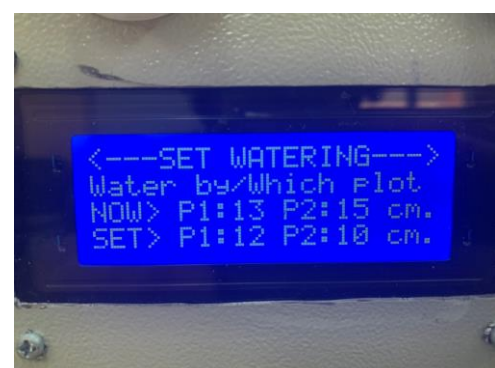


Figure 4.32 Plot's watering settings main page



Figure 4.33 Watering function page



Figure 4.34 Main display of liquid filling into mixer tank



Figure 4.35 Tank/plots selecting page



Figure 4.36 Tank filling functions page



Figure 4.37 Main page of fertilizer recipe

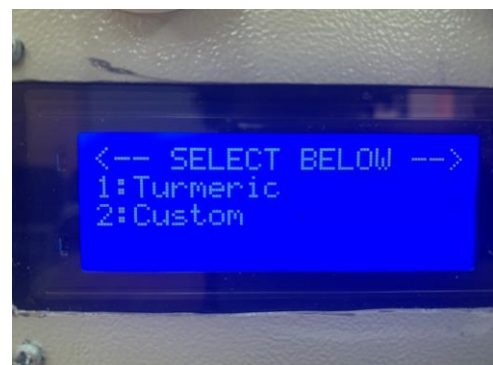


Figure 4.38 Selecting recipe page

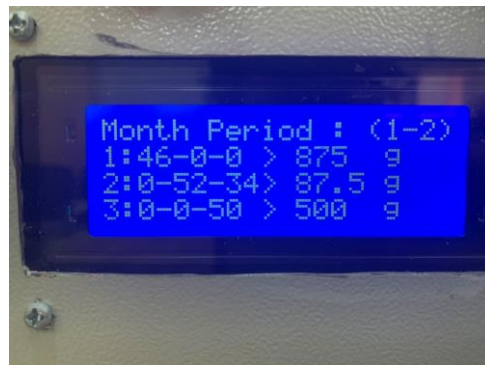


Figure 4.39 Selecting month period of built-in fertilizer page



Figure 4.40 Custom recipe page



Figure 4.41 pH settings page



Figure 4.42 EC offset settings page



Figure 4.43 Peristaltic pump hose cleaning page

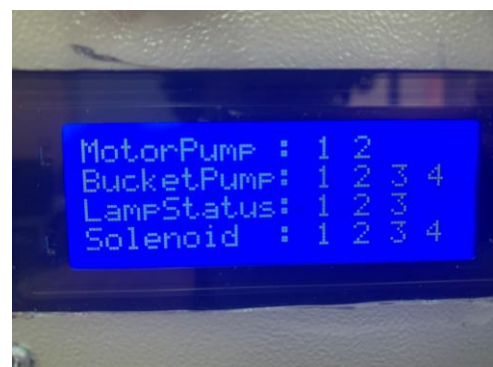


Figure 4.44 Operation testing of all essential devices page



Figure 4.45 Date/time setting page

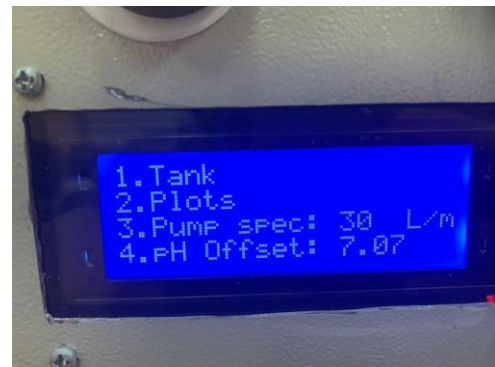


Figure 4.46 Technical settings page



Figure 4.47 Technical settings page
(Cont'd)



Figure 4.48 Factory restoration
notification display

Display of task operations:



Figure 4.49 Page of selecting
fertilizer barrel to fill into the mixer
tank



Figure 4.50 Fertilizer filling progress
display



Figure 4.51 Display progress of filling liquid into the mixer tank



Figure 4.52 Display progress of filling liquid into the plot

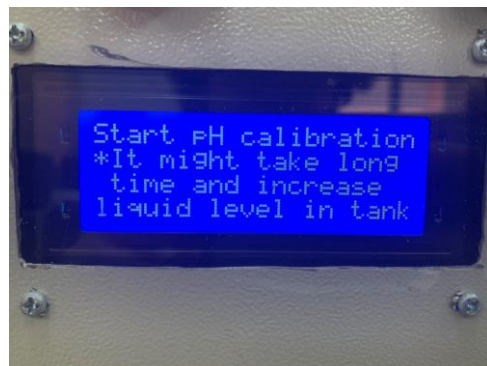


Figure 4.53 Display of pH calibration notification

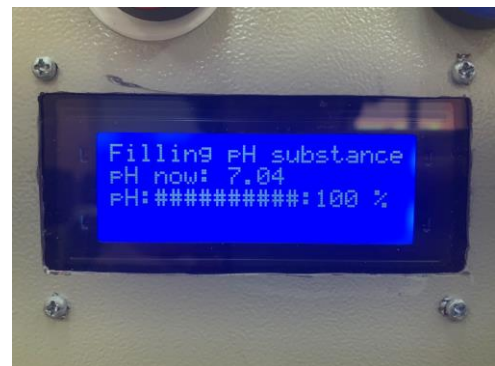


Figure 4.54 Display progress of pH calibration



Figure 4.55 Another display of pH calibration notification



Figure 4.56 Display of looping liquid in mixer tank



Figure 4.57 Display progress and status of plots watering



Figure 4.58 Display of cancelling automation task



Figure 4.59 Display progress of autonomous function to filling fertilizers into the mixer tank



Figure 4.60 Display progress of autonomous function to filling water into the mixer tank

4.2.7 Objectives review

Result: From table 4.3 This machine can meet to all objectives of project.

No.	Objective lists	Yes	No
1	User can select working mode (Auto / Hybrid / Manual)	✓	
2	User can choose operation of the subsystem.	✓	
3	User can choose liquid to be filled into the mixing tank.	✓	
4	User can set the liquid level in the mixer tank.	✓	
5	User can select liquid to water into plots.	✓	
6	User can set the water level into plots.	✓	
7	User can define pH value in mixing tank.	✓	
8	User can monitor and calibrate EC value in plots.	✓	
9	User can set the recipes of NPK fertilizer.	✓	
10	User can design their own fertilizer recipe.	✓	
11	User can set time of the machine.	✓	
12	User can test all essential devices in the system.	✓	
13	User can set technical constants of the system.	✓	
14	User can restore all setting and constant back to factory settings.	✓	

Table 4.3 Check list of objectives

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

The Intelligent watering system achieved the objective goal and gives an impressive result. The system can perform well mixing fertilizer ingredients accurately. The sensor work with no problems and the system is performing as expected. The safety also exists to prevent unexpected process that might occur during operation such as short circuit or motor overloading. The safety also prevents the system form breaking down.

During developing the system, we faced with power problem and noise from electronics components. But after fixing the power supply and adding electronic noise cancelling part such as snubber circuit, surge suppressor, diode, and opto-isolated relay the system can perform in a long period of time without any problems.

The system helps reduce a good amount of time from watering and fertilizing plants and waste the comes from human error. The system performs accurately and can be used in greenhouses, but the system needs to be checked and repair be for use to get the best performance from it.

5.2 Discussion

The intelligent watering system has a lot of functions to cover all need for plants, But the watering system is still cannot be used for plants that were planted in soil. This system only supports hydro-phonic type of planting because the system does not measure the moisture in the plot instead it measures the level of water in the plot. However, to have the system be supported for soil planting, the system will have to be designed and developed purposely for it.

In the system, there is still no button or faction to control the motors and solenoid directly or manually. This function can allow users to use the system even when the MCU is not working properly. This can be added in the future but is just not added to the system yet.

This system cannot be further more than two meters from the sensors because the signal can be lost in the wire. That can cause calculating analog signal to digital inaccurate before sending the signal to the control unit such as Modbus or 4-20mA.

pH sensor and EC sensor that is used in this system is still not stable. The sensor gives unstable output when the water is not still or when there is a lot amount of water in the tank. This leads to problem of finding the exact number which will have to be dealt with in the future.

This system is designed to be used in the location or area that accessing WIFI is not possible. That means controlling the system remotely is not possible. To develop the system to be controlled remotely, the system needs the module that can access internet or micro controller that can be connected to internet to connect via serial communication with Arduino mega 2560 that drives the system. Also, Lora-wan technology will likely be the answer for this system.

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REFERENCES

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APPENDICES

APPENDIX A

EEPROM

The microcontroller on the Arduino AVR based board has EEPROM: memory whose values are kept when the board is turned off (like a tiny hard drive). This library enables you to read and write those bytes but it have a limited number of writing (about 100,000 times)

The supported micro-controllers on the various Arduino boards have different amounts of EEPROM: 1024 bytes on the ATmega328P, 512 bytes on the ATmega168 and ATmega8, 4 KB (4096 bytes) on the ATmega1280 and ATmega2560. The Arduino boards have an emulated EEPROM space of 1024 bytes.

To use this library

```
#include <EEPROM.h>
```

Examples

EEPROM.Clear: Clear the bytes in the EEPROM.

EEPROM.Read: Read the EEPROM and send its values to the computer.

EEPROM.Write: Stores values from an analog input to the EEPROM.

EEPROM.Crc: Calculates the CRC of EEPROM contents as if it was an array.

EEPROM.Get: Get values from EEPROM and prints as float on serial.

EEPROM.Iteration: Understand how to go through the EEPROM memory locations.

EEPROM.Put: Put values in EEPROM using variable semantics.

EEPROM.Update: Stores values read from A0 into EEPROM, writing the value only if different, to increase EEPROM life.

APPENDIX B

SNUBBER CIRCUIT

These circuits are very useful in handling the spikes of voltage that arise from the operation of inductive loads. And we use it many point in the systems.

RC snubbers

A simple RC snubber uses a small resistor (R) in series with a small capacitor (C). This combination can be used to suppress the rapid rise in voltage across a thyristor, preventing the erroneous turn-on of the thyristor; it does this by limiting the rate of rise in voltage (dV/dt) across the thyristor to a value which will not trigger it. An appropriately-designed RC snubber can be used with either DC or AC loads. This sort of snubber is commonly used with inductive loads such as electric motors. The voltage across a capacitor cannot change instantaneously, so a decreasing transient current will flow through it for a small fraction of a second, allowing the voltage across the switch to increase more slowly when the switch is opened. Determination of voltage rating can be difficult owing to the nature of transient waveforms, and may be defined simply by the power rating of the snubber components and the application. RC snubbers can be made discretely and are also built as a single component.

Diode snubbers

Main article: Flyback diode

When the current flowing is DC, a simple rectifier diode is often employed as a snubber. The snubber diode is wired in parallel with an inductive load (such as a relay coil or electric motor). The diode is installed so that it does not conduct under normal conditions. When the external driving current is interrupted, the inductor current flows instead through the diode. The stored energy of the inductor is then gradually dissipated by the diode voltage drop and the resistance of the inductor itself. One disadvantage of using a simple rectifier diode as a snubber is that the diode allows current to continue flowing for some time, causing the inductor to remain active for slightly longer than desired. When such a snubber is utilized in a relay, this effect may cause a significant delay in the drop out, or disengagement, of the actuator.

The diode must immediately enter into forward conduction mode as the driving current is interrupted. Most ordinary diodes, even "slow" power silicon diodes, are able to turn on very quickly, in contrast to their slow reverse recovery time. These are sufficient for snubbing electromechanical devices such as relays and motors.

In high-speed cases, where the switching is faster than 10 nanoseconds, such as in certain switching power regulators, "fast", "ultrafast", or Schottky diodes may be required.

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